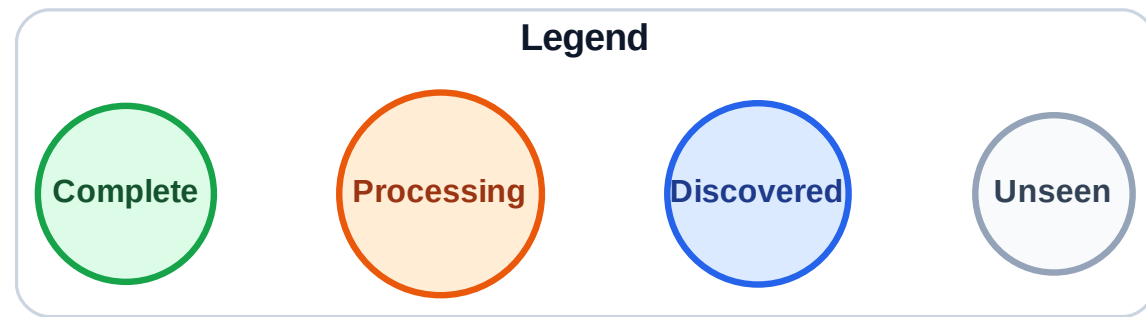
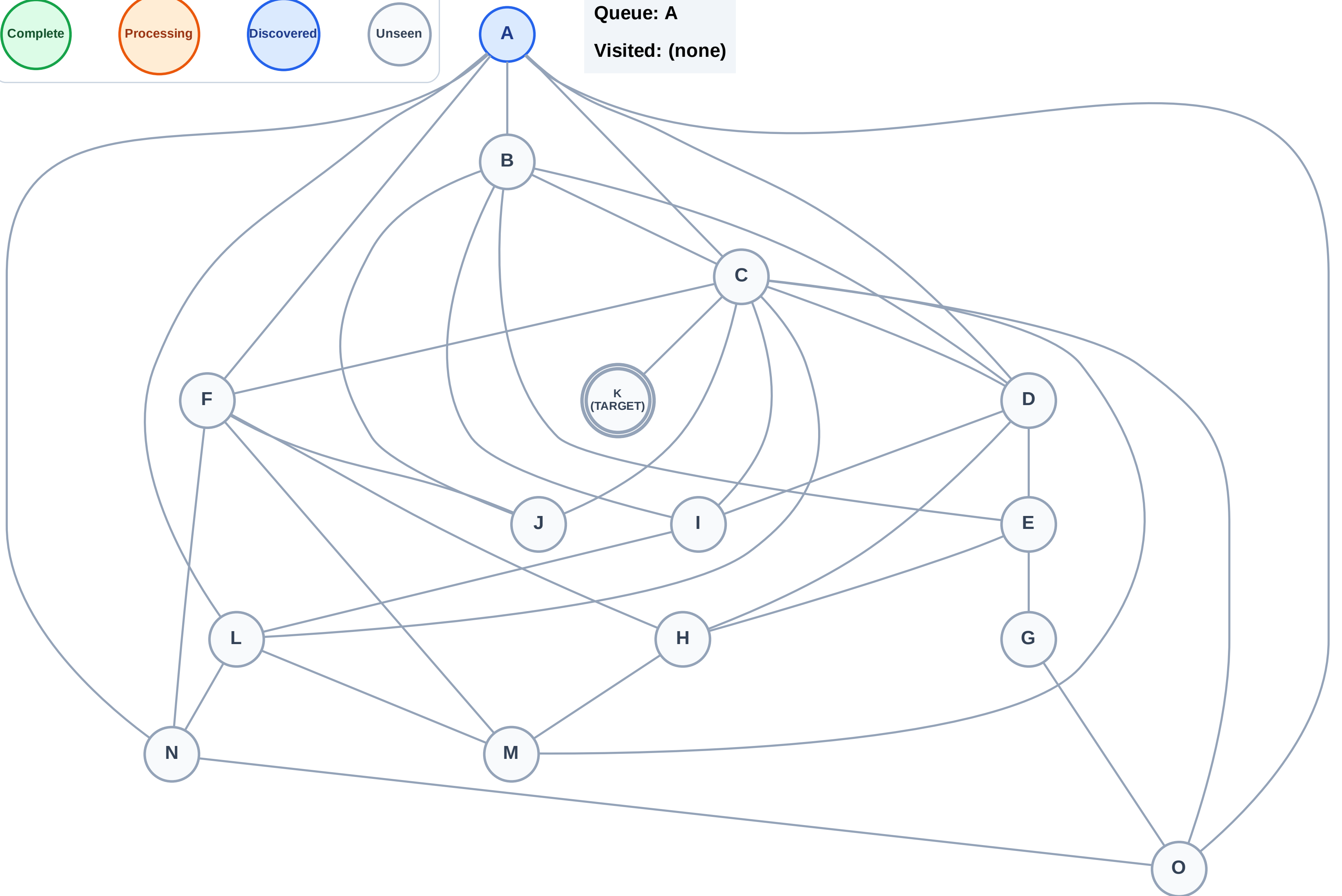


BFS Traversal

Step 1: Start at A (Seeking target node K)

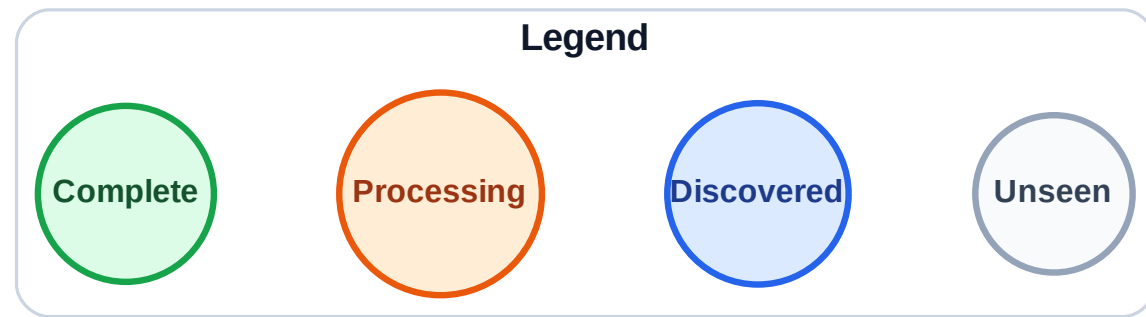


Queue: A
Visited: (none)



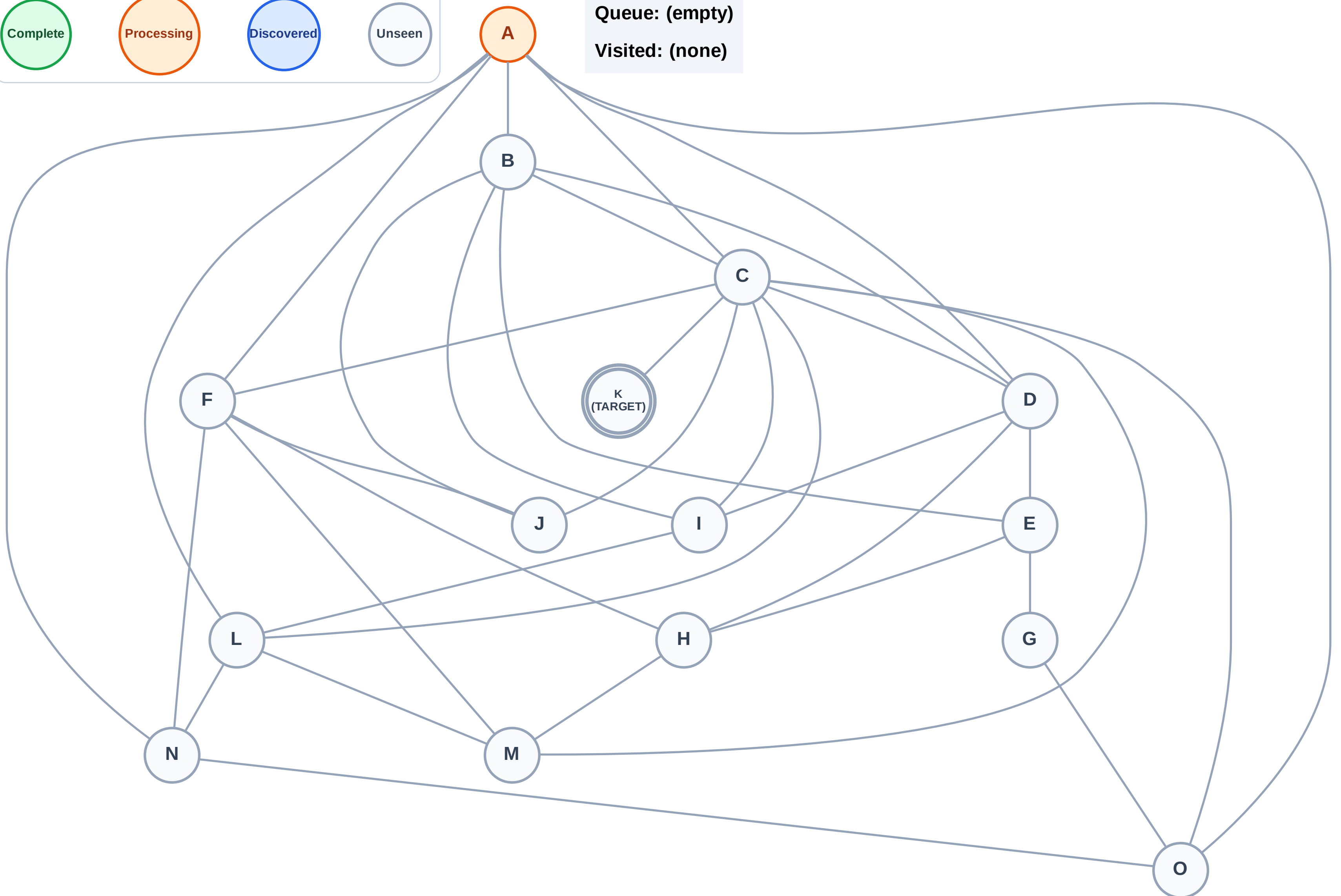
BFS Traversal

Step 2: Process node A



Queue: (empty)

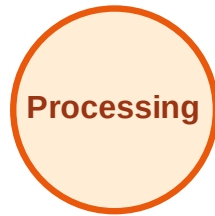
Visited: (none)



BFS Traversal

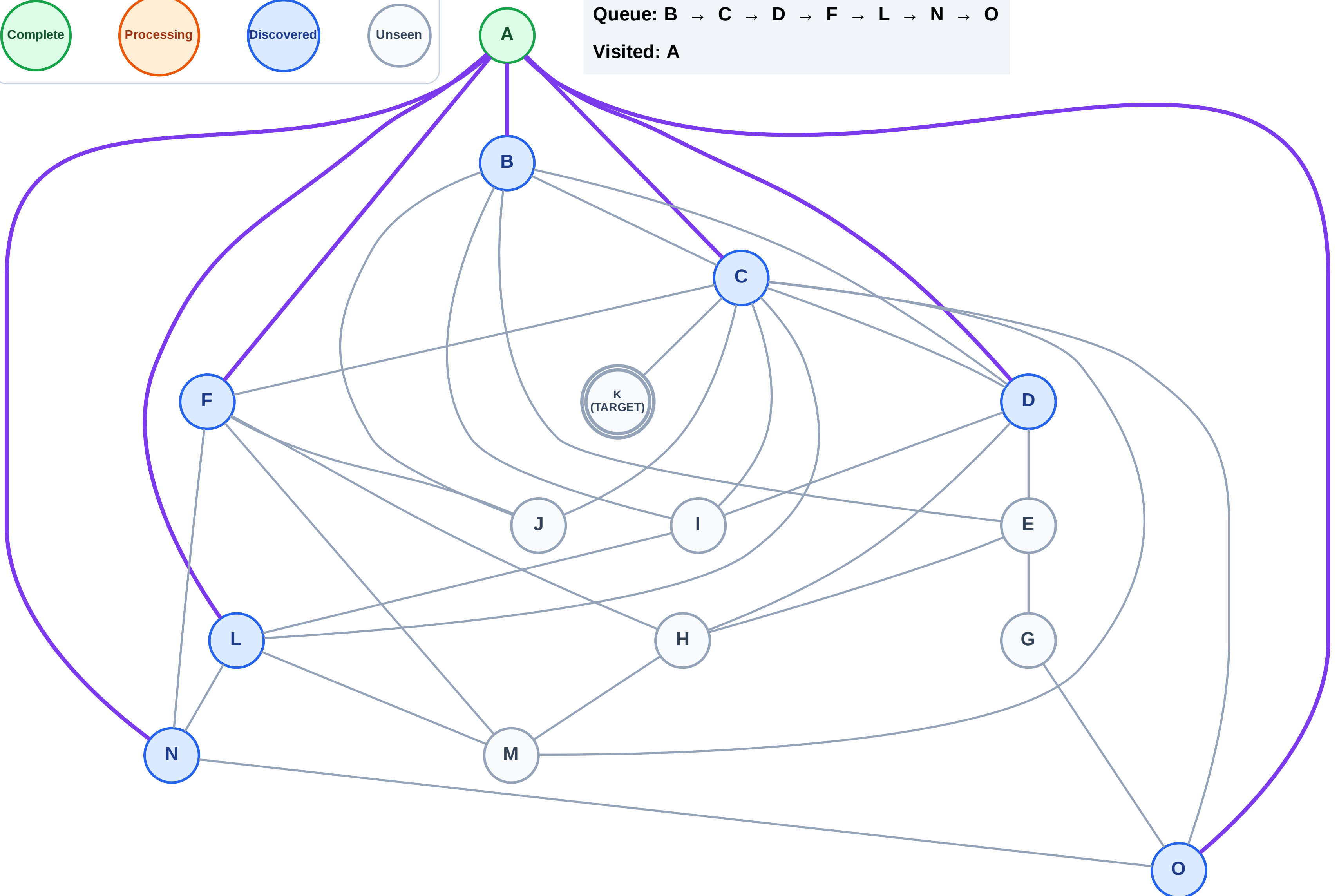
Step 3: Discover B, C, D, F, L, N, O from A

Legend



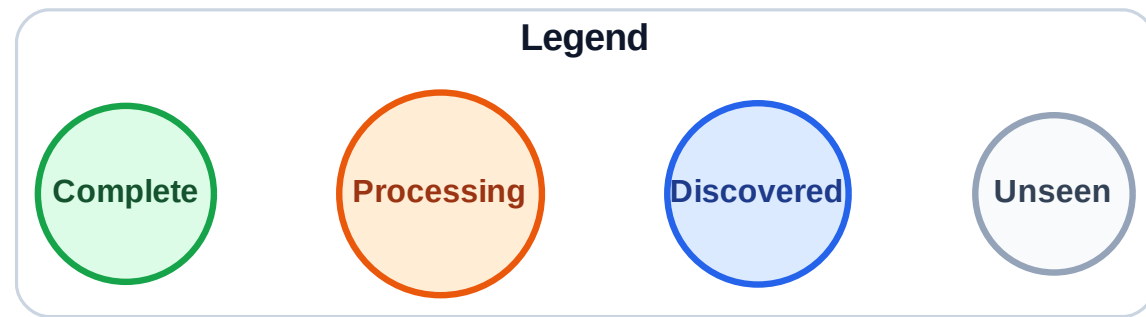
Queue: B → C → D → F → L → N → O

Visited: A



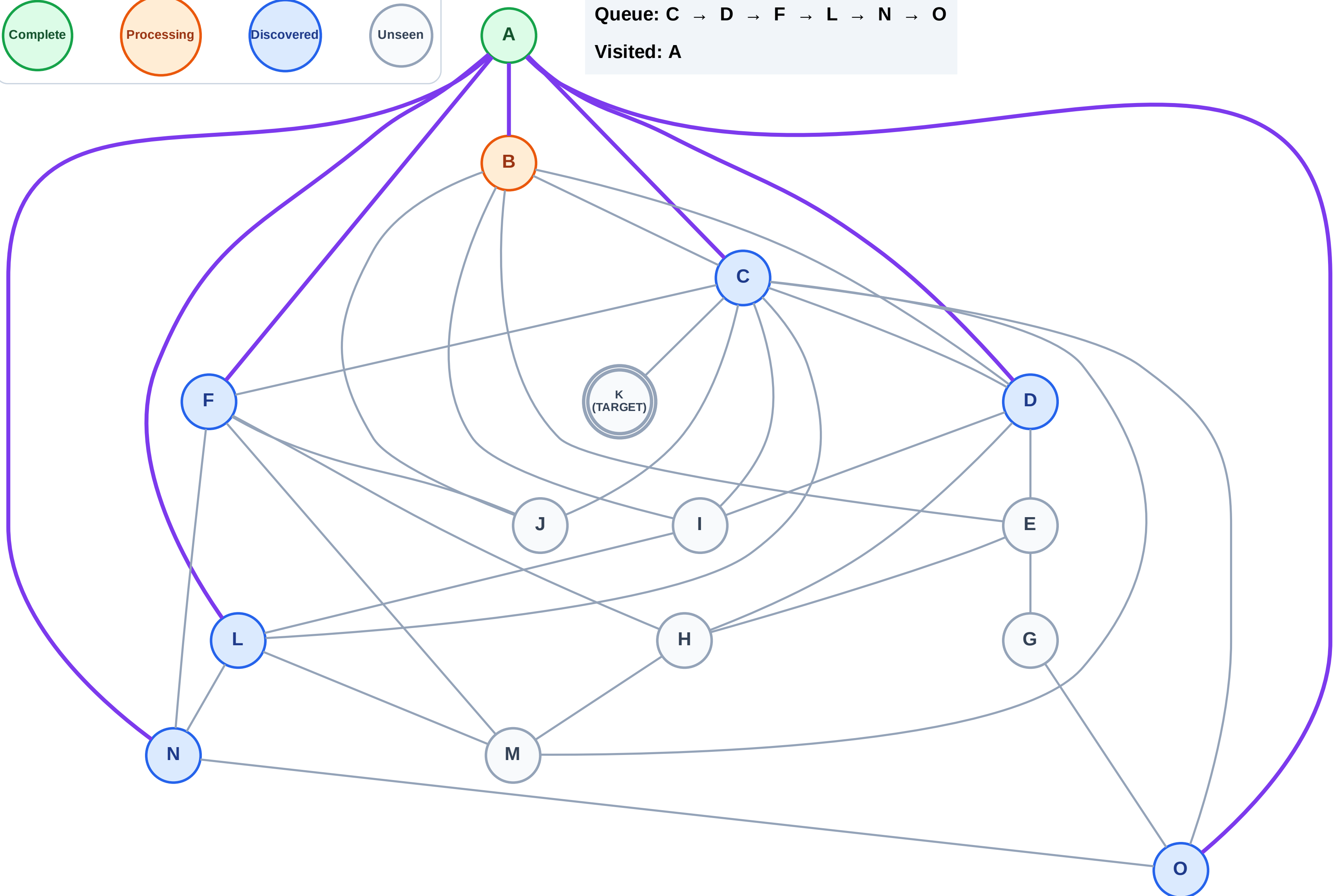
BFS Traversal

Step 4: Process node B



Queue: C → D → F → L → N → O

Visited: A



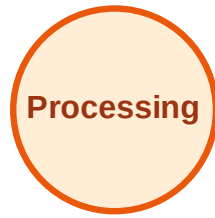
BFS Traversal

Step 5: Discover E, I, J from B

Legend



Complete



Processing



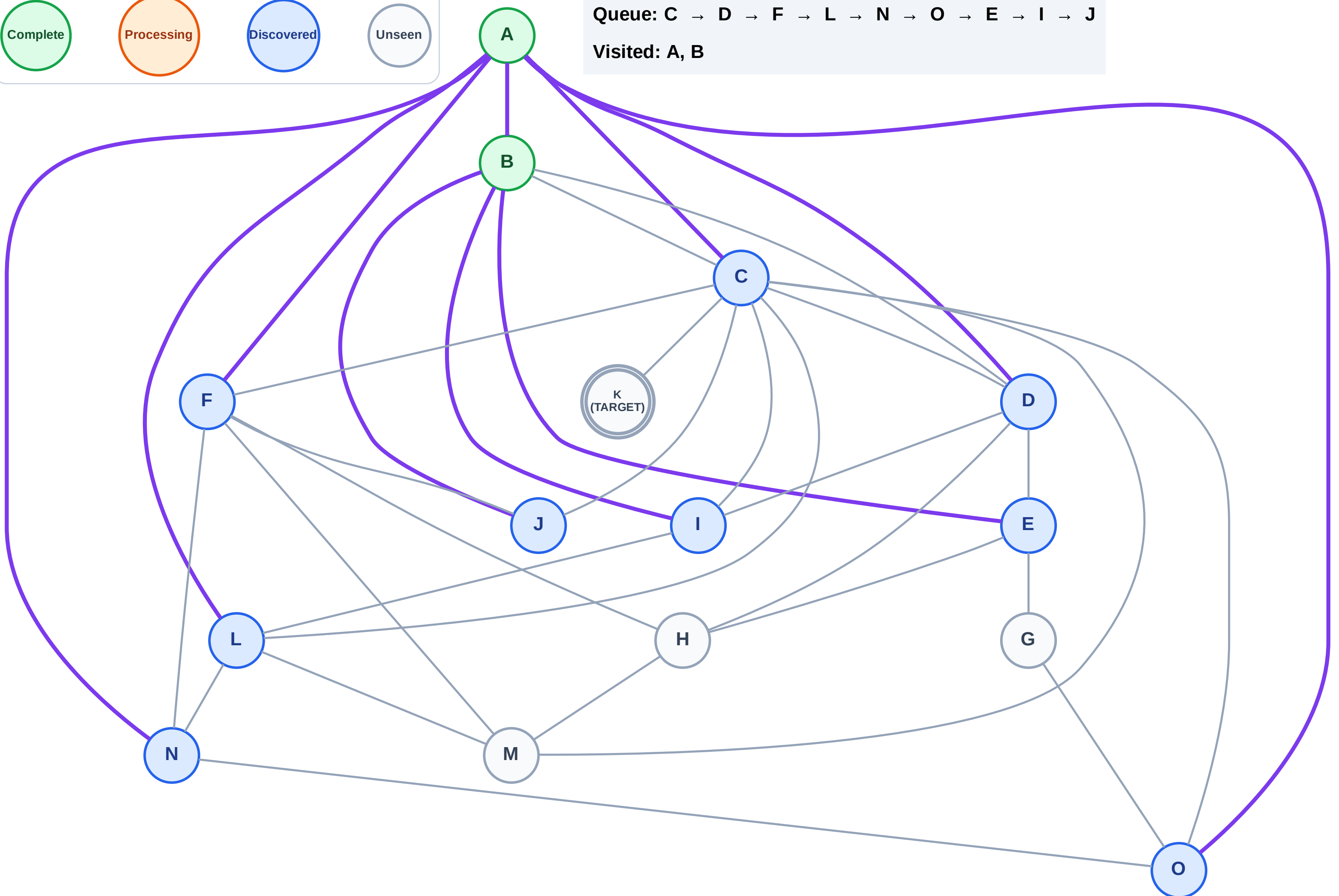
Discovered



Unseen

Queue: C → D → F → L → N → O → E → I → J

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

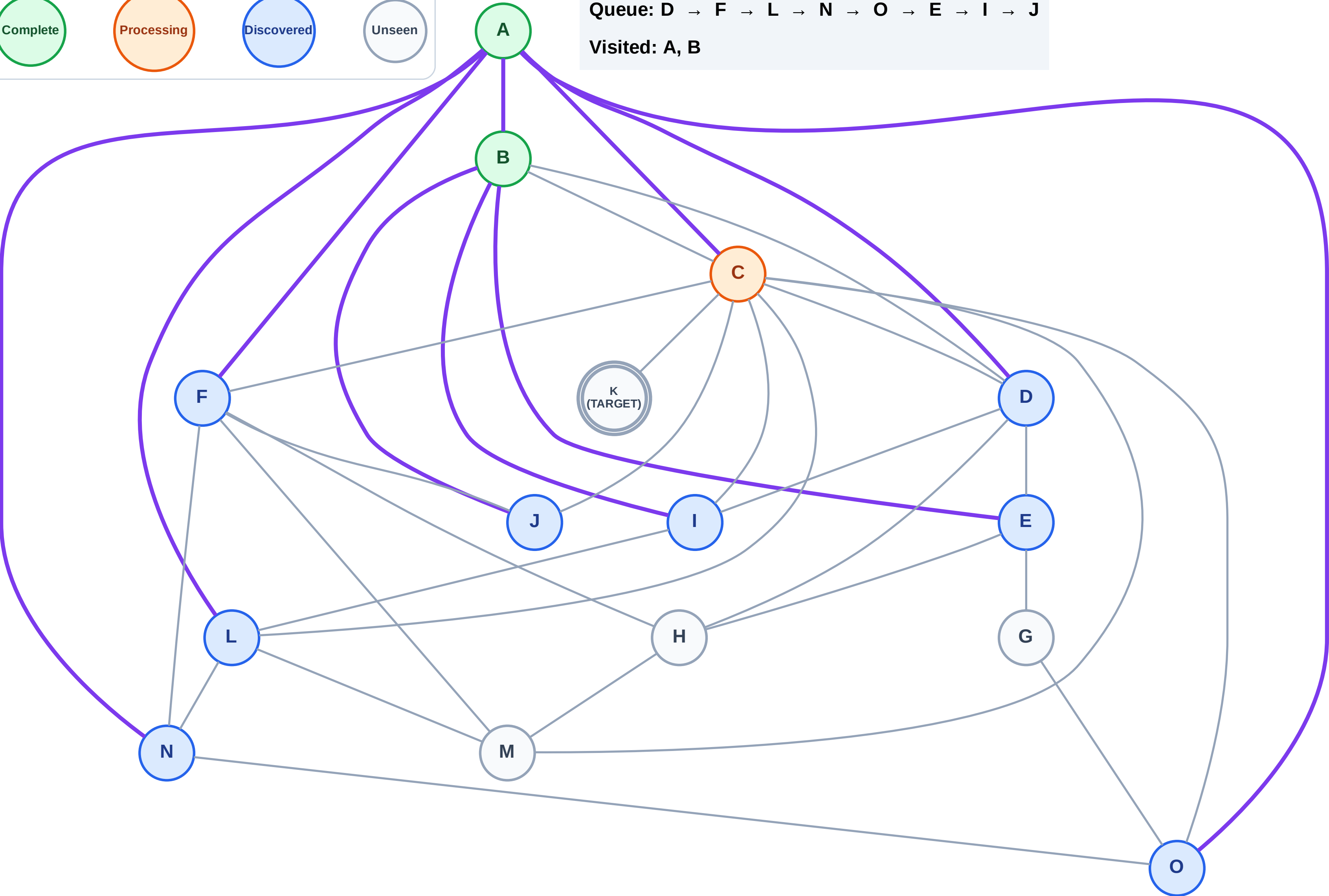
Processing

Discovered

Unseen

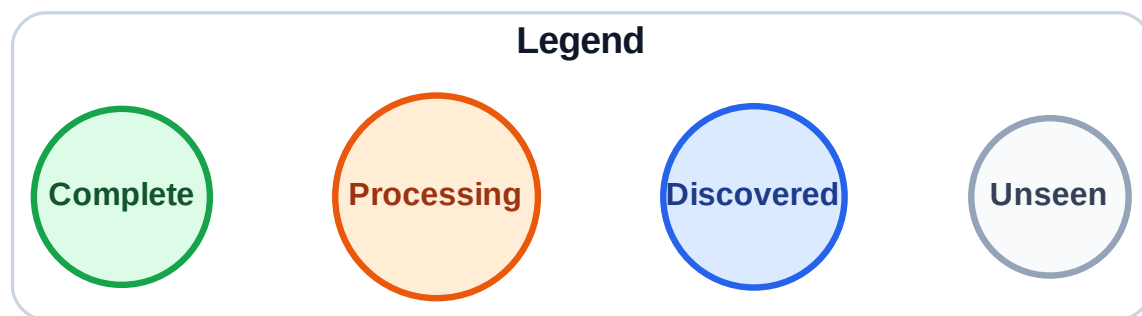
Queue: D → F → L → N → O → E → I → J

Visited: A, B



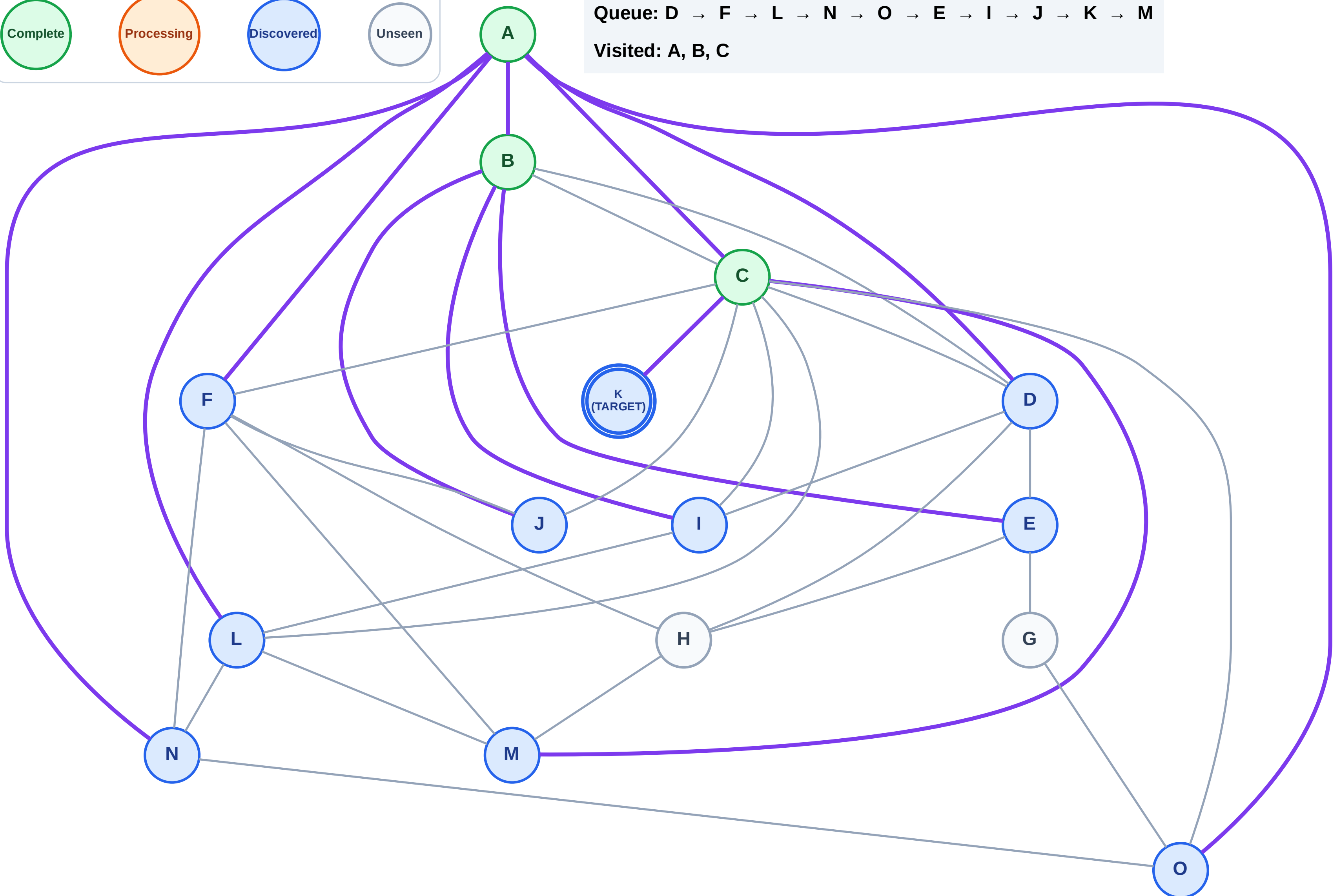
BFS Traversal

Step 7: Discover K, M from C



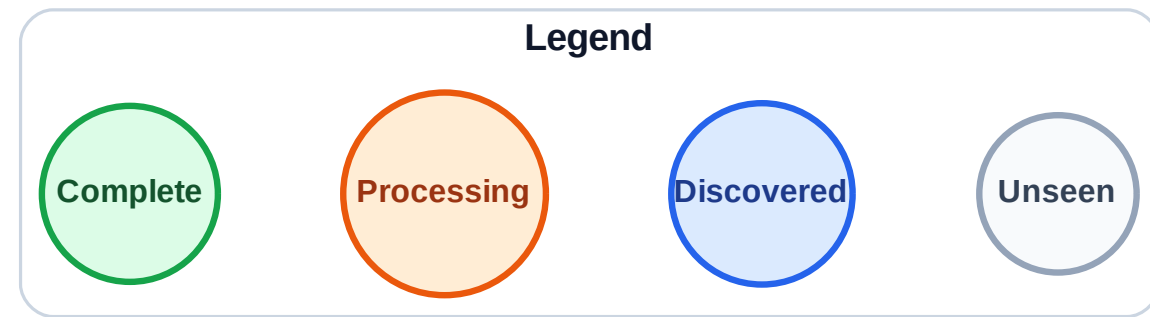
Queue: D → F → L → N → O → E → I → J → K → M

Visited: A, B, C



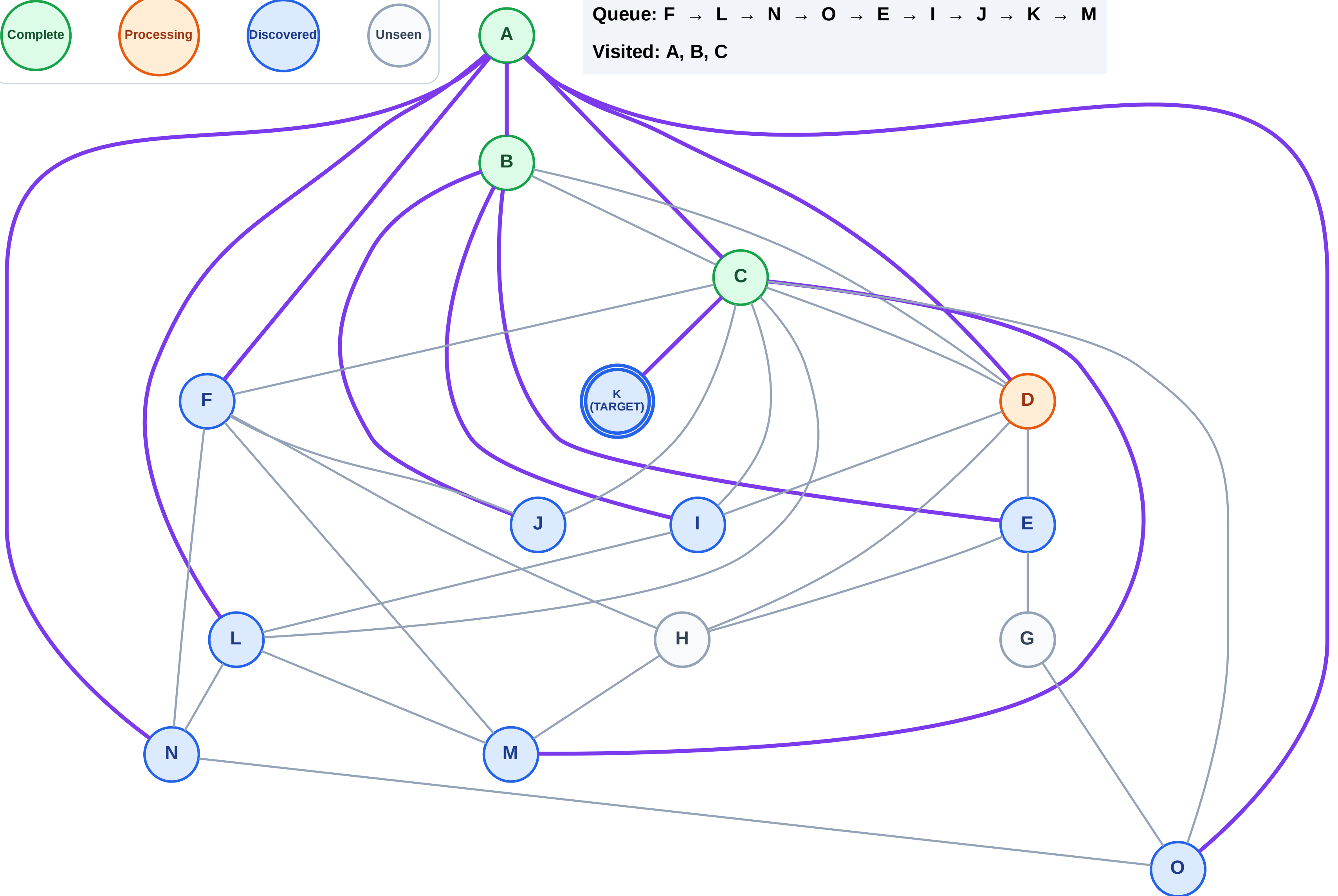
BFS Traversal

Step 8: Process node D



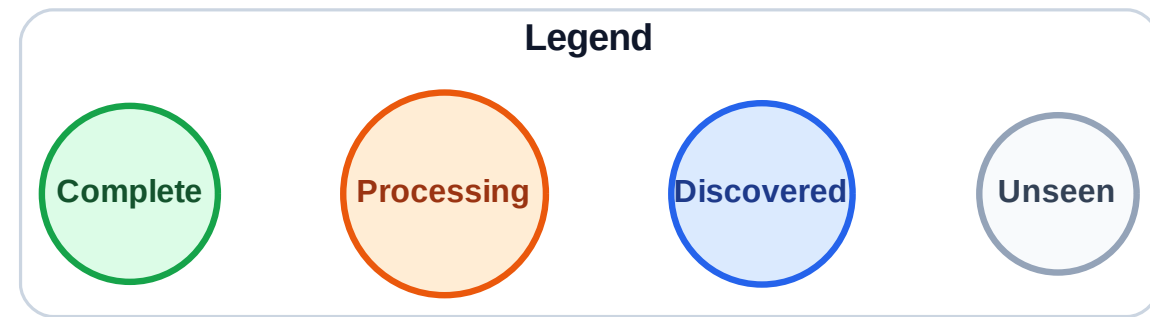
Queue: F → L → N → O → E → I → J → K → M

Visited: A, B, C



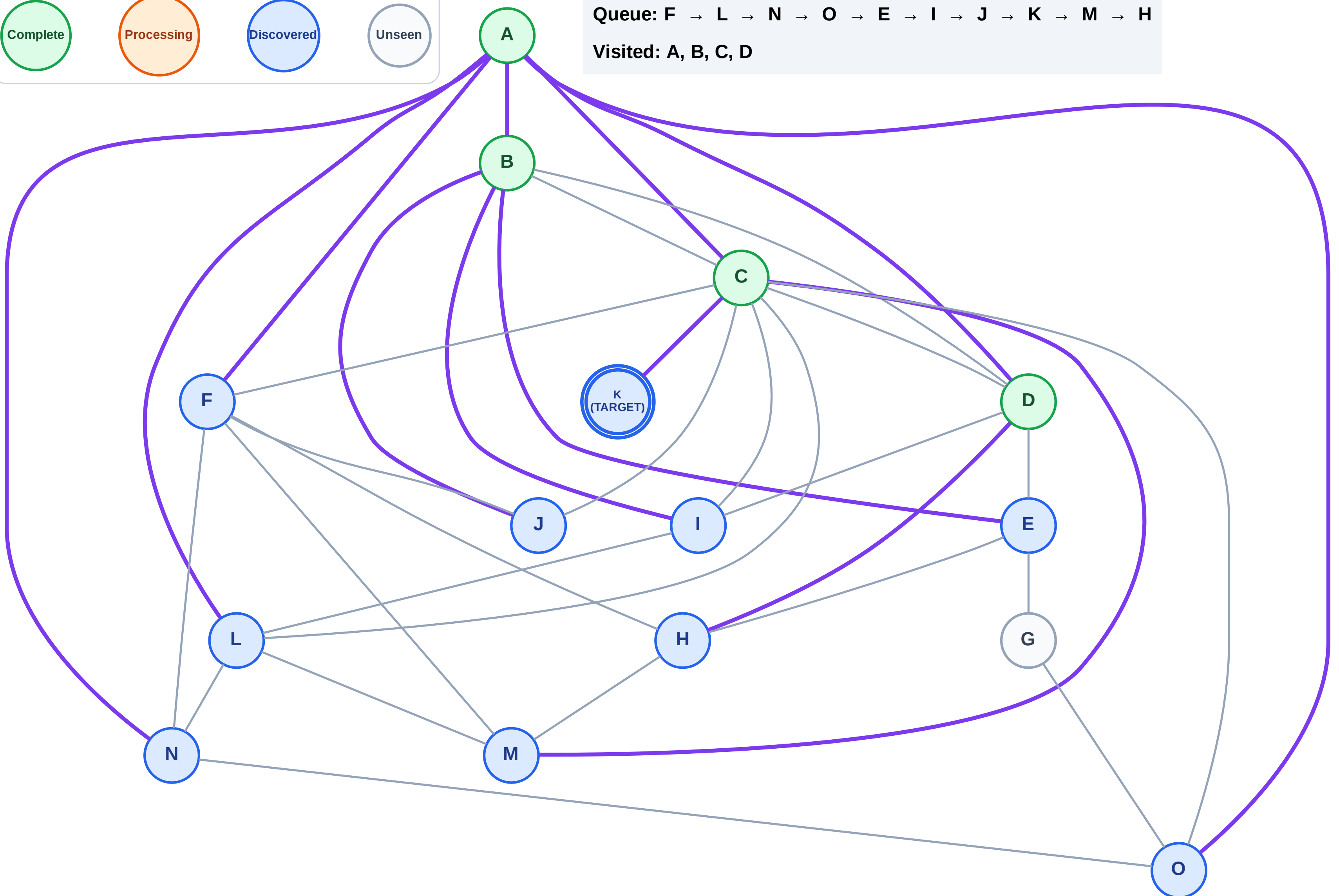
BFS Traversal

Step 9: Discover H from D



Queue: F → L → N → O → E → I → J → K → M → H

Visited: A, B, C, D



BFS Traversal

Step 10: Process node F

Legend

Complete

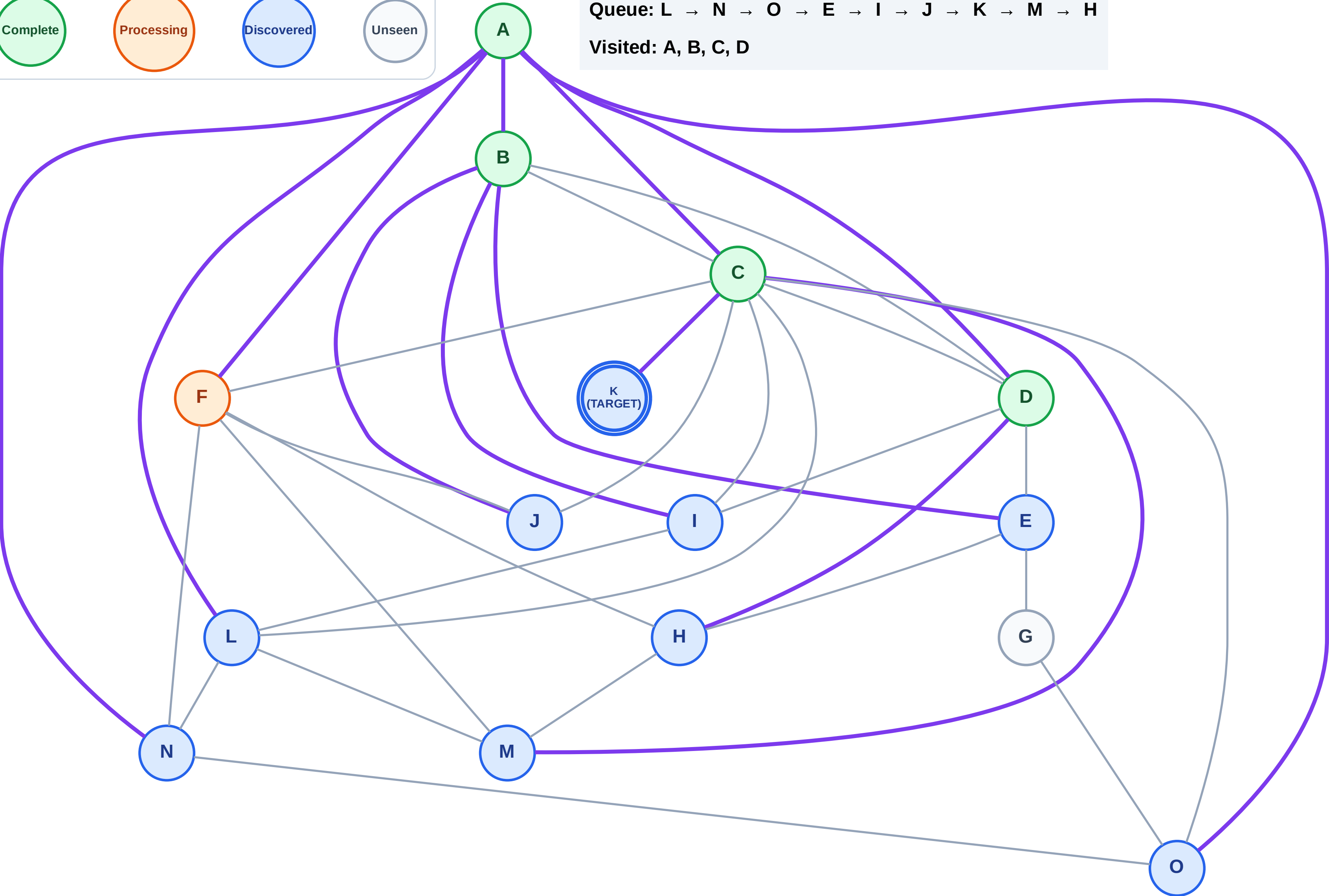
Processing

Discovered

Unseen

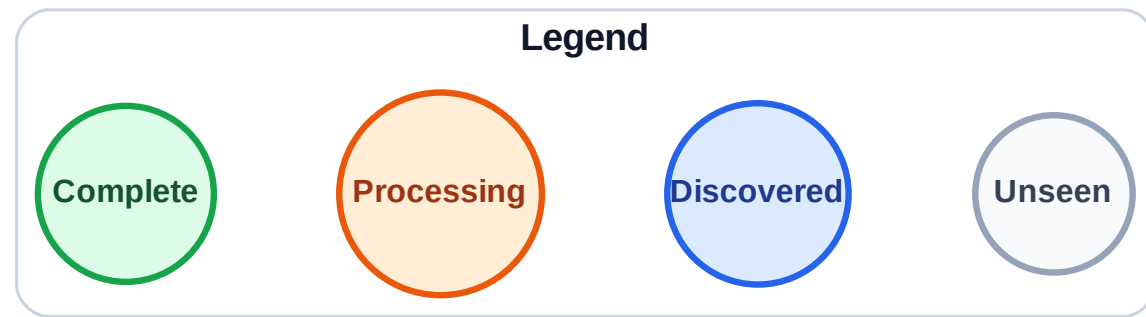
Queue: L → N → O → E → I → J → K → M → H

Visited: A, B, C, D



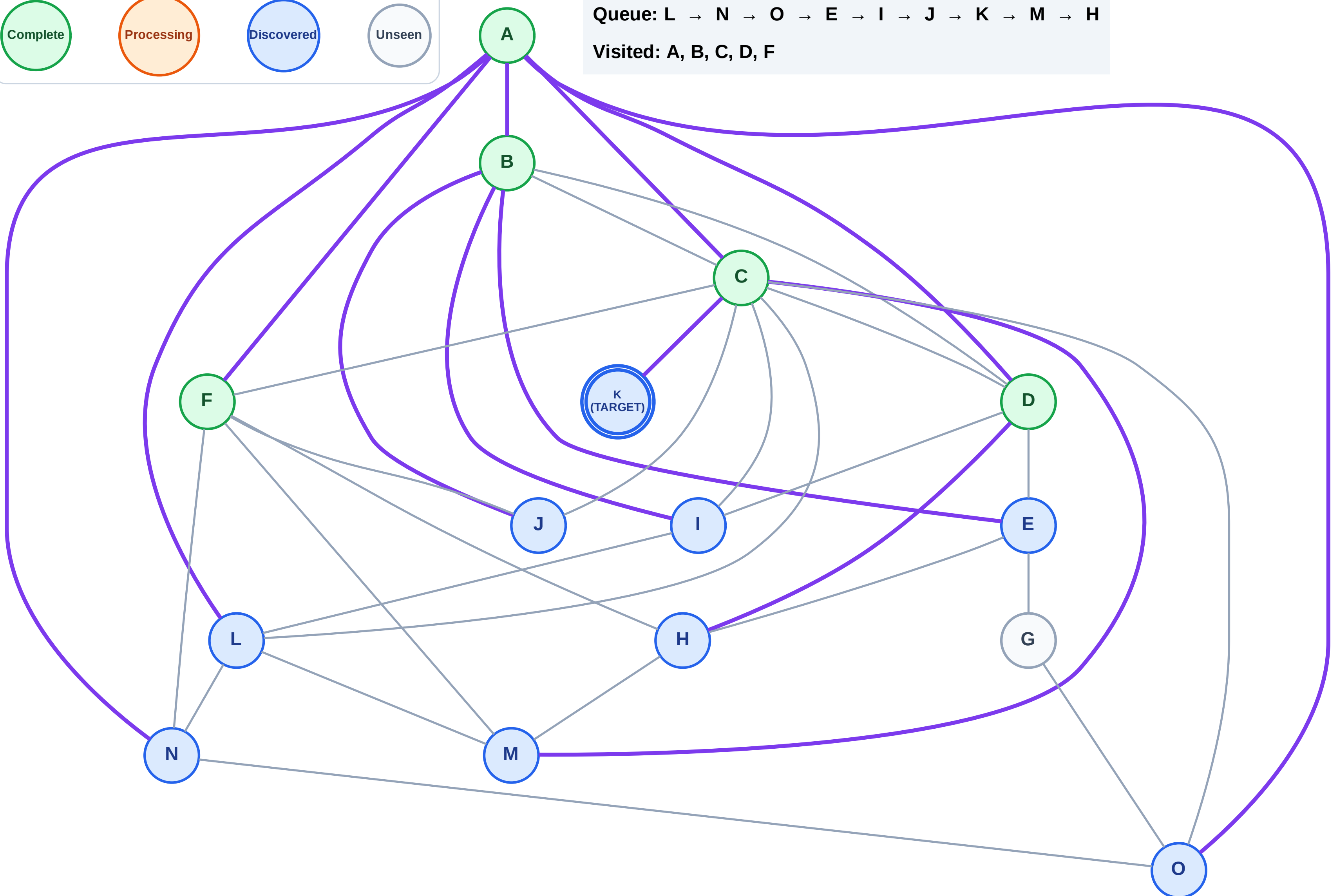
BFS Traversal

Step 11: No new neighbors from F



Queue: L → N → O → E → I → J → K → M → H

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node L

Legend

Complete

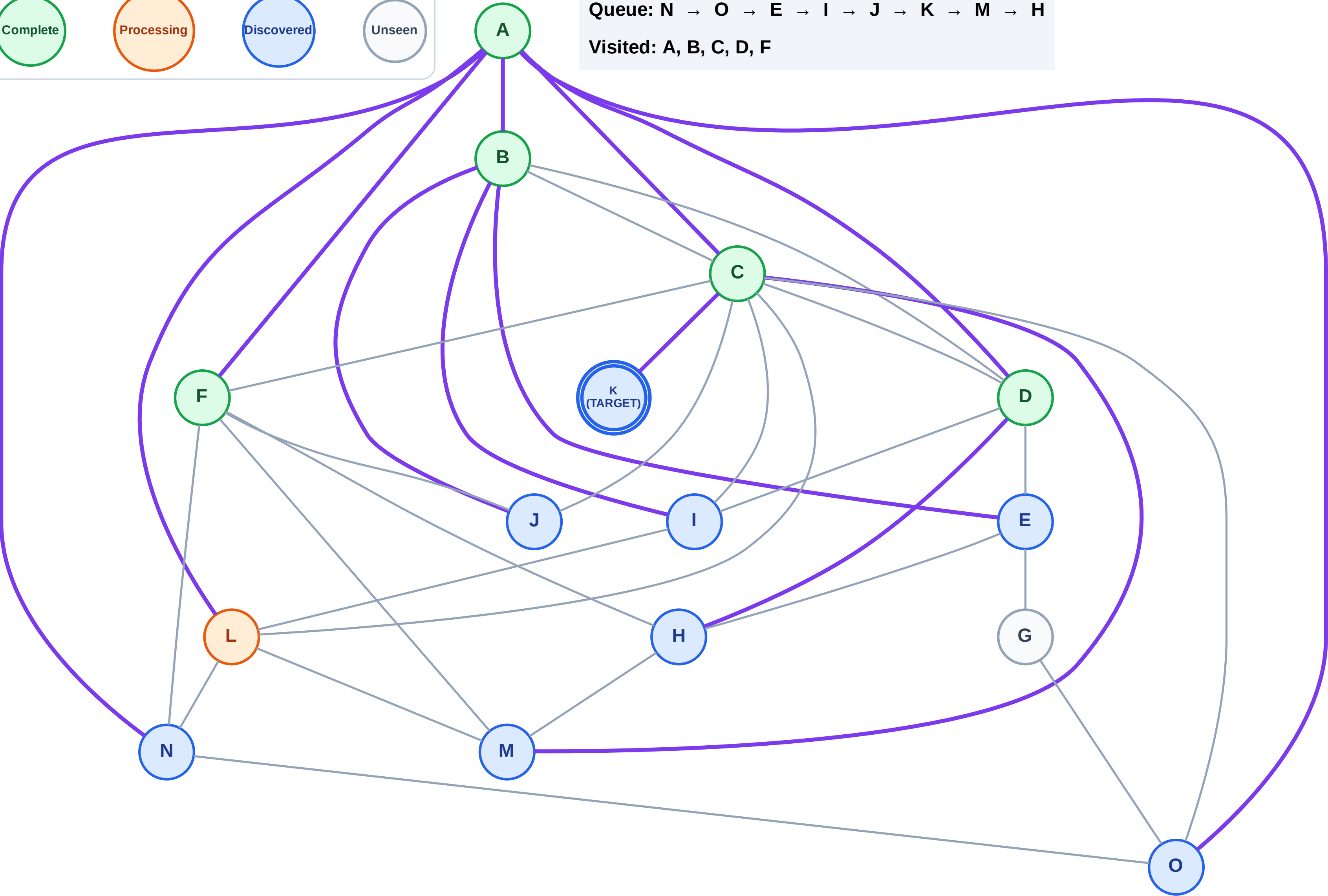
Processing

Discovered

Unseen

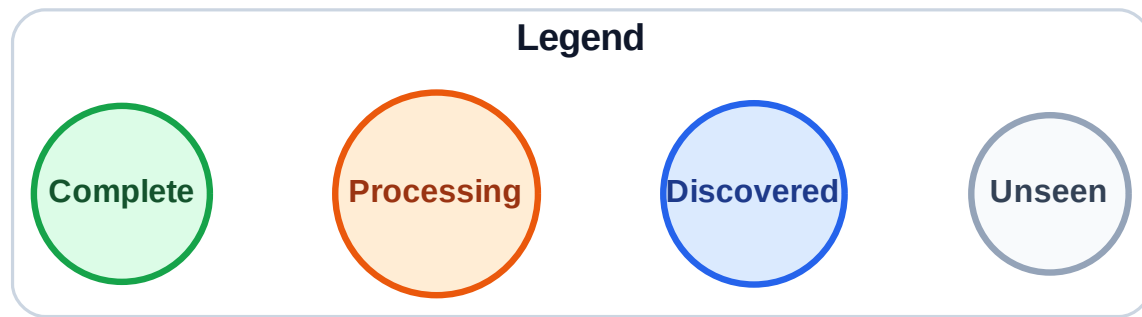
Queue: N → O → E → I → J → K → M → H

Visited: A, B, C, D, F



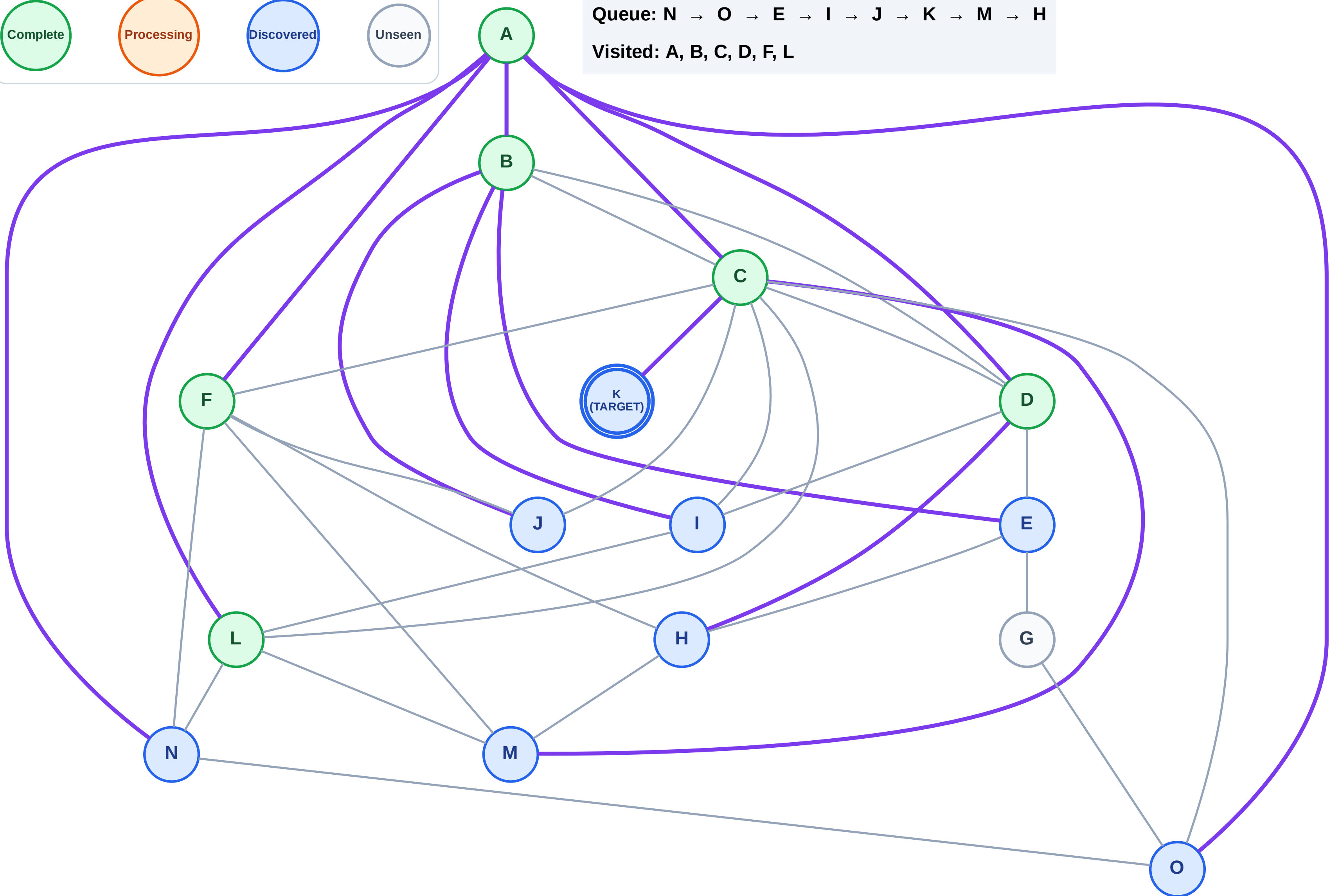
BFS Traversal

Step 13: No new neighbors from L



Queue: N → O → E → I → J → K → M → H

Visited: A, B, C, D, F, L



BFS Traversal

Step 14: Process node N

Legend

Complete

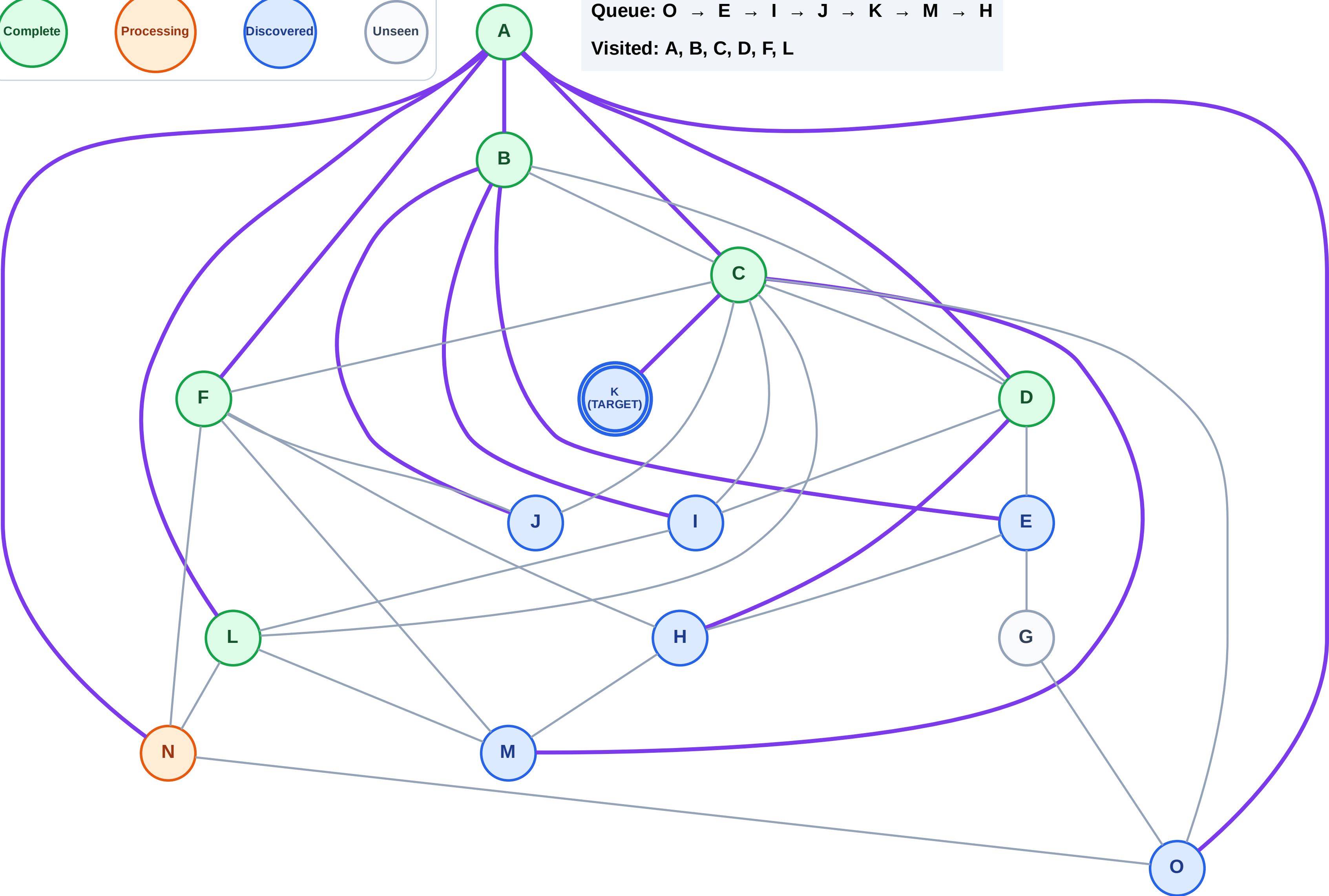
Processing

Discovered

Unseen

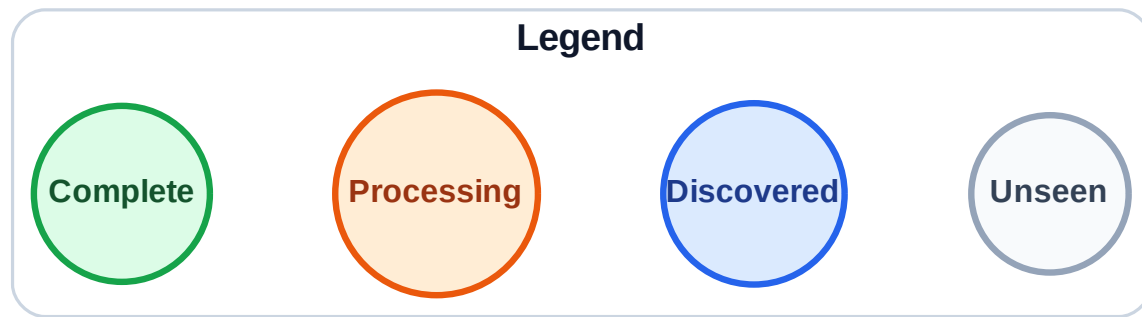
Queue: O → E → I → J → K → M → H

Visited: A, B, C, D, F, L



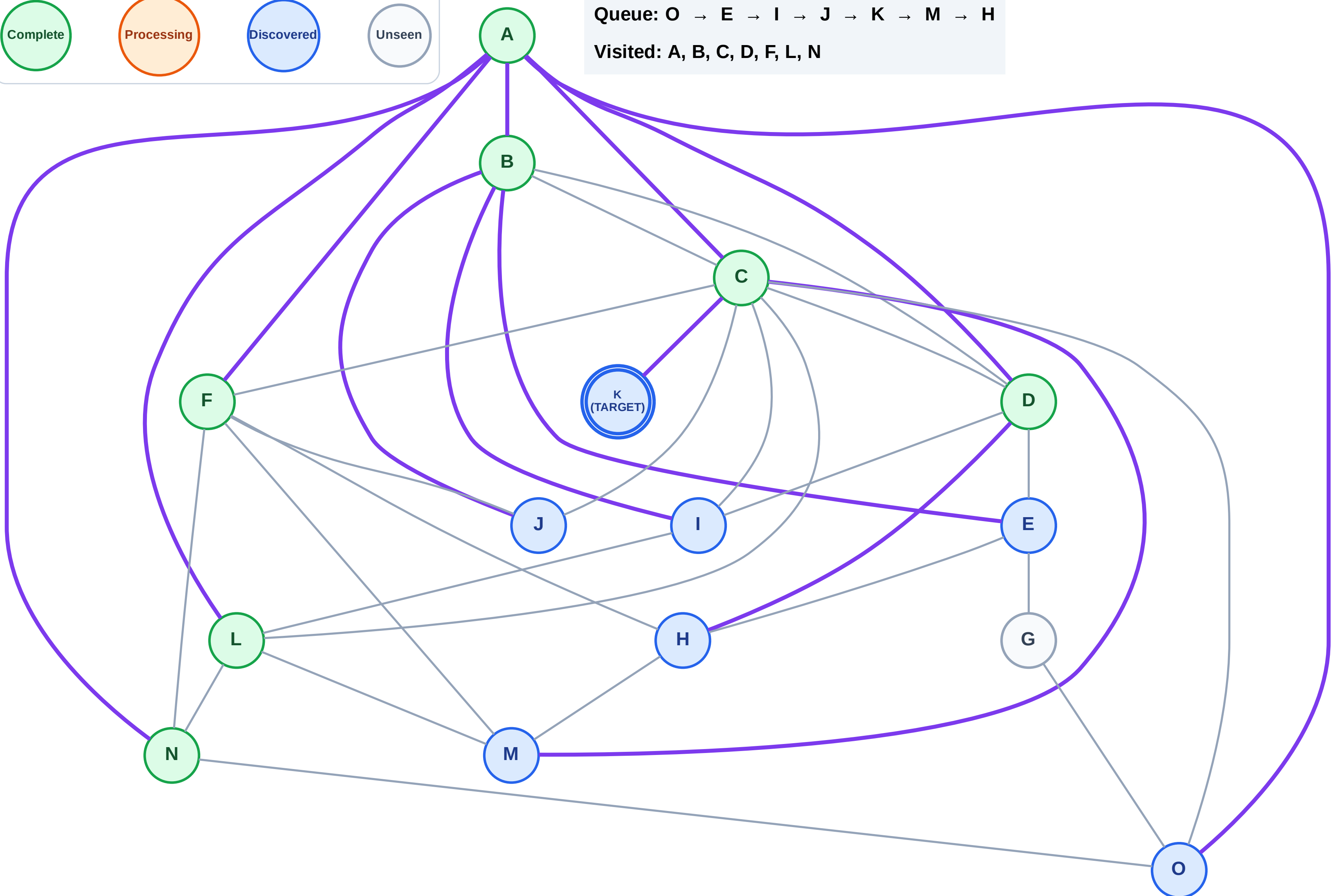
BFS Traversal

Step 15: No new neighbors from N



Queue: O → E → I → J → K → M → H

Visited: A, B, C, D, F, L, N



BFS Traversal

Step 16: Process node O

Legend

Complete

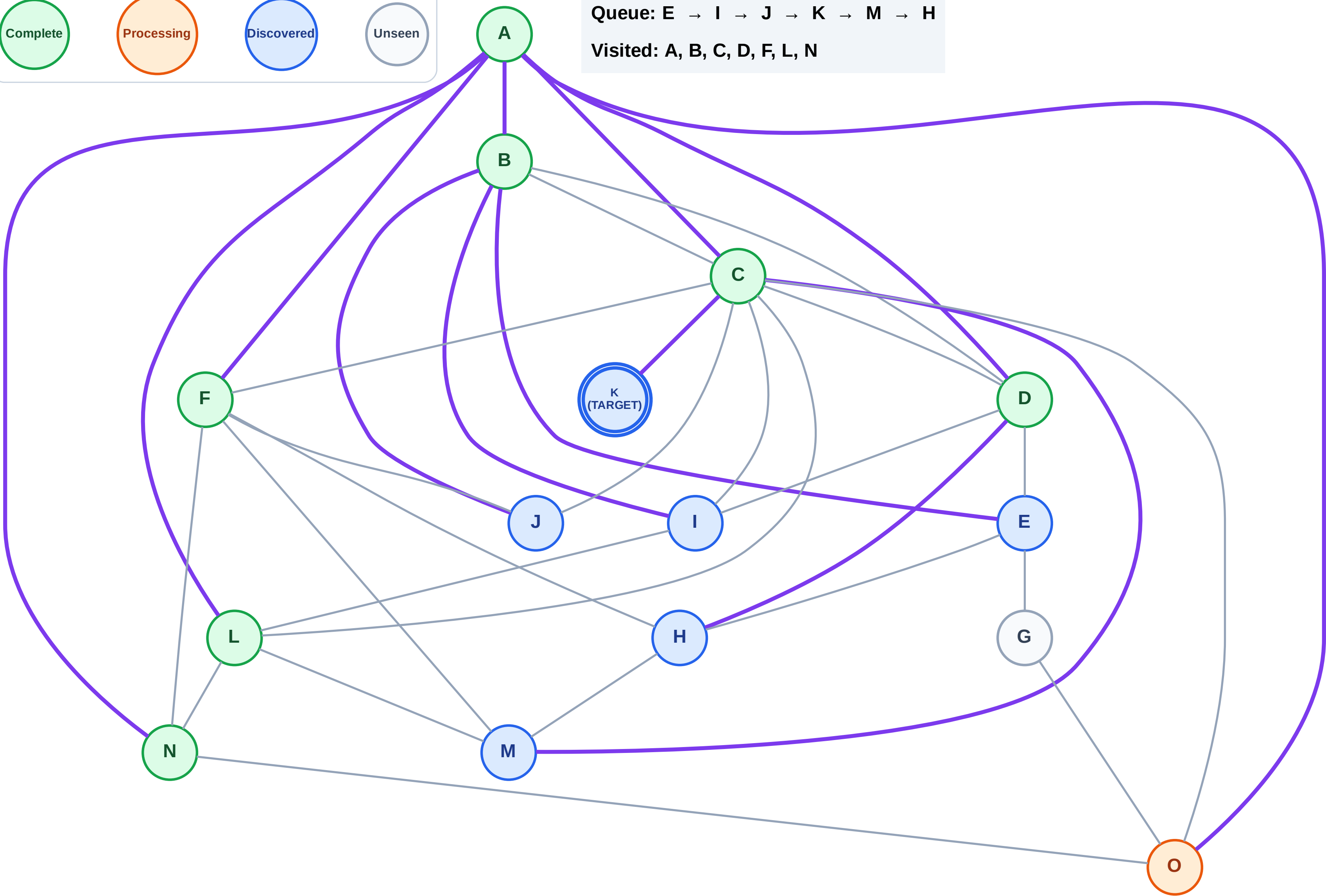
Processing

Discovered

Unseen

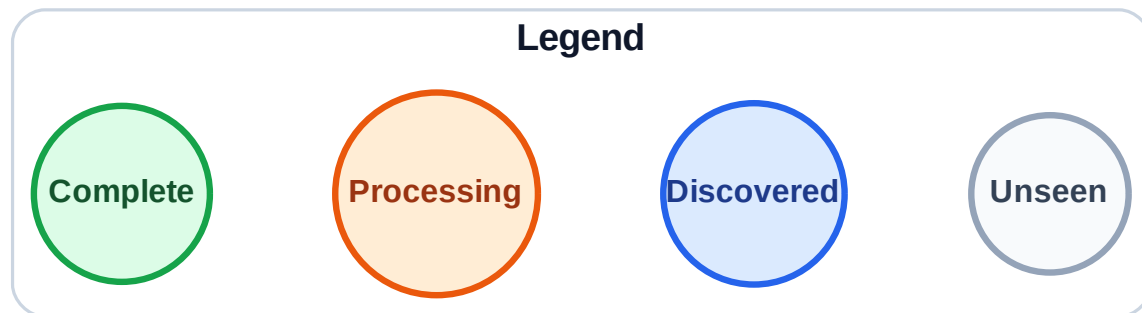
Queue: E → I → J → K → M → H

Visited: A, B, C, D, F, L, N



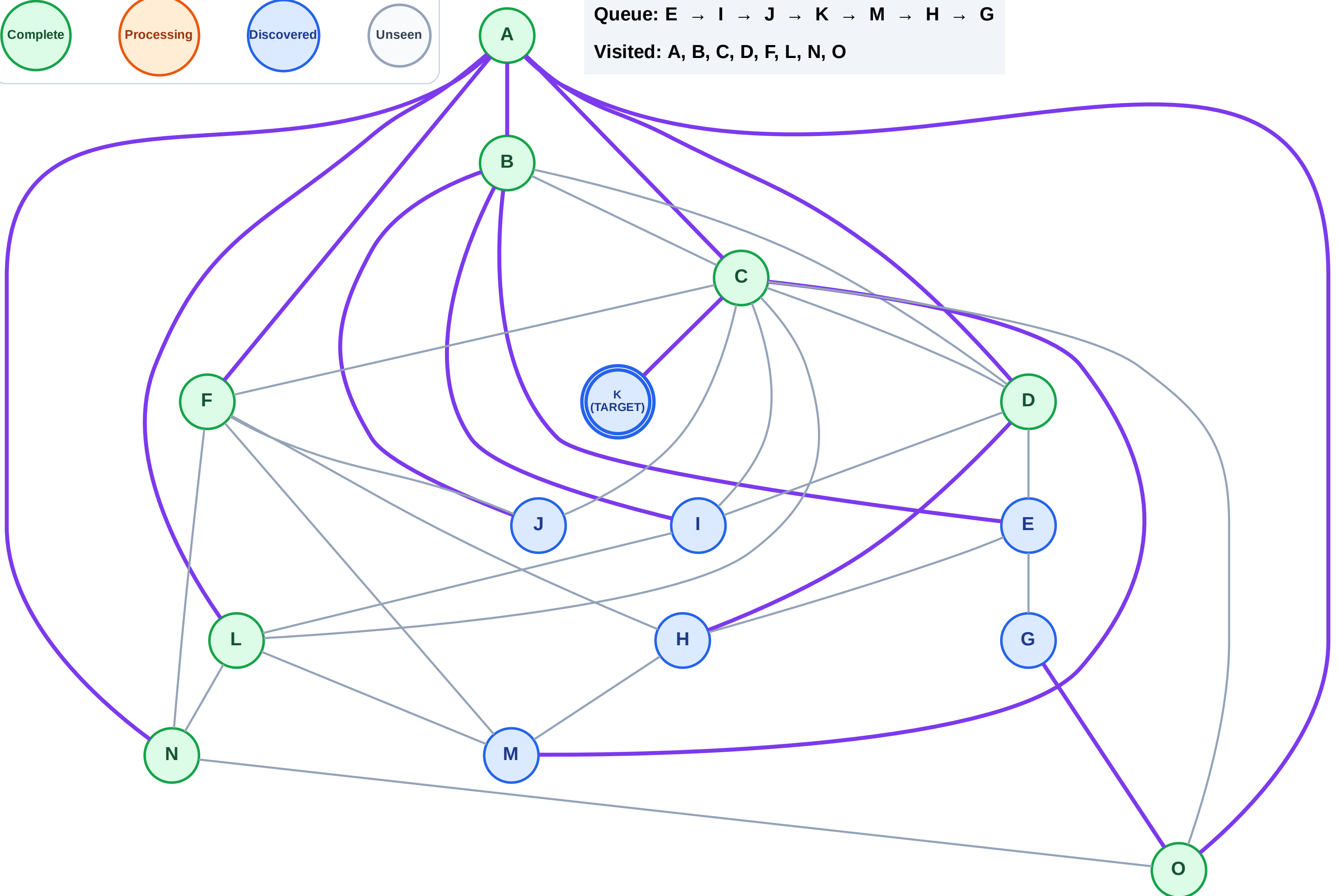
BFS Traversal

Step 17: Discover G from O



Queue: E → I → J → K → M → H → G

Visited: A, B, C, D, F, L, N, O



BFS Traversal

Step 18: Process node E

Legend

Complete

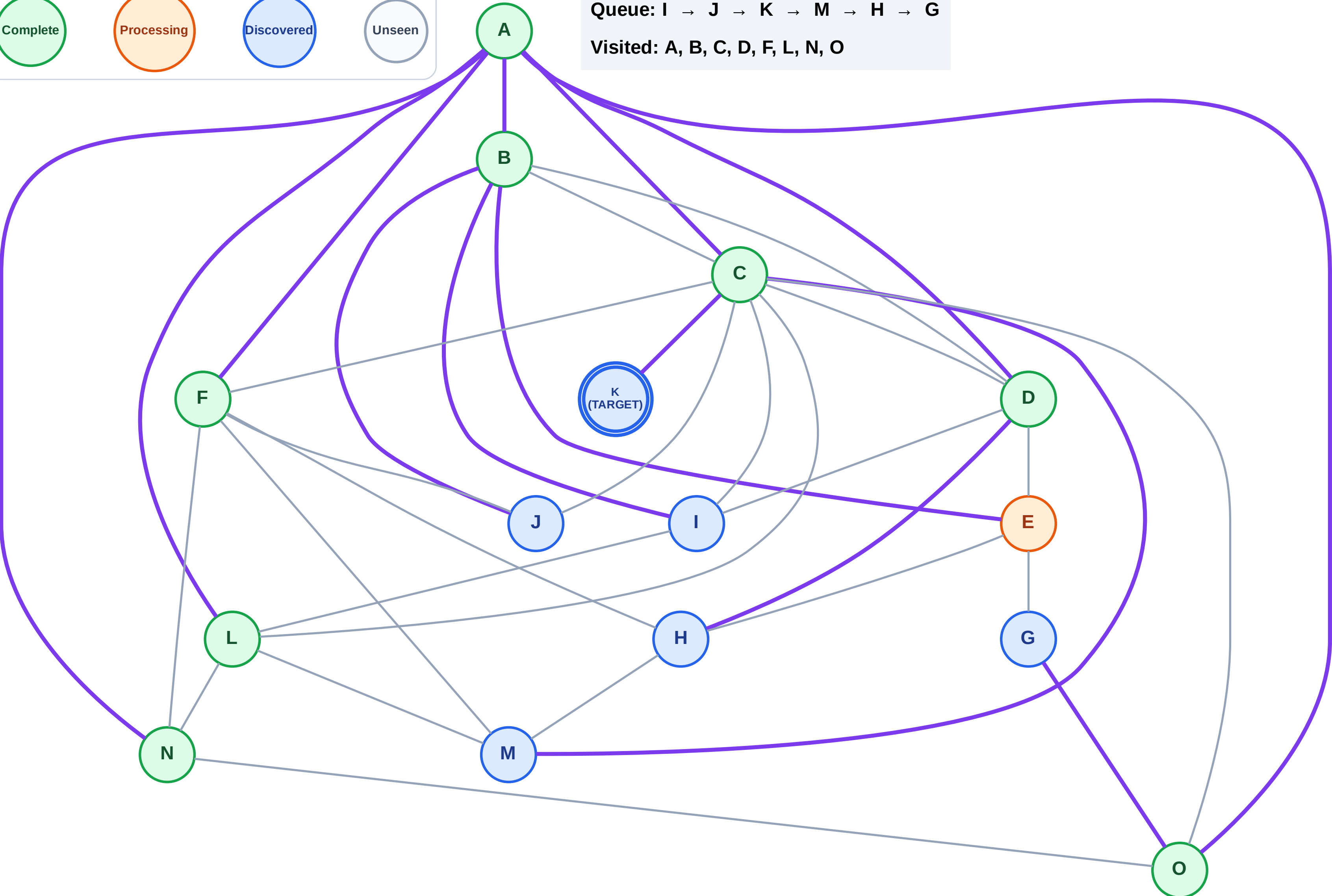
Processing

Discovered

Unseen

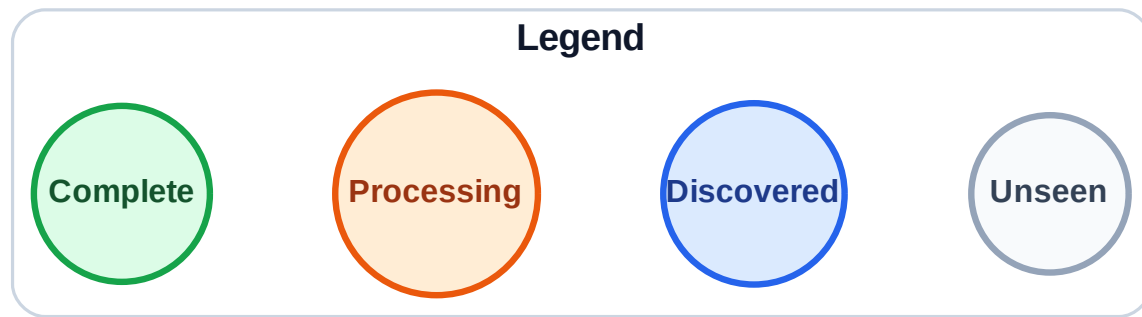
Queue: I → J → K → M → H → G

Visited: A, B, C, D, F, L, N, O



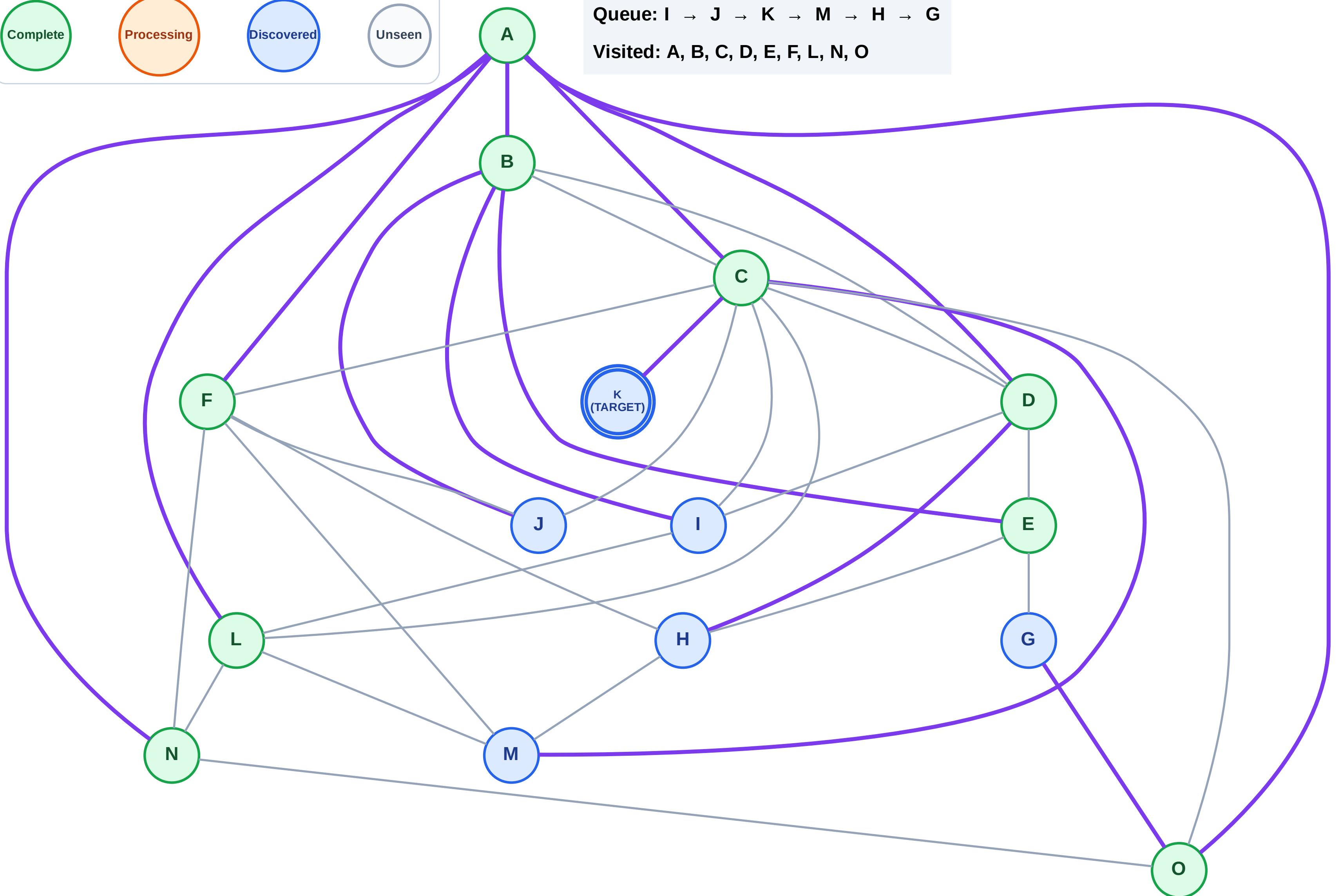
BFS Traversal

Step 19: No new neighbors from E



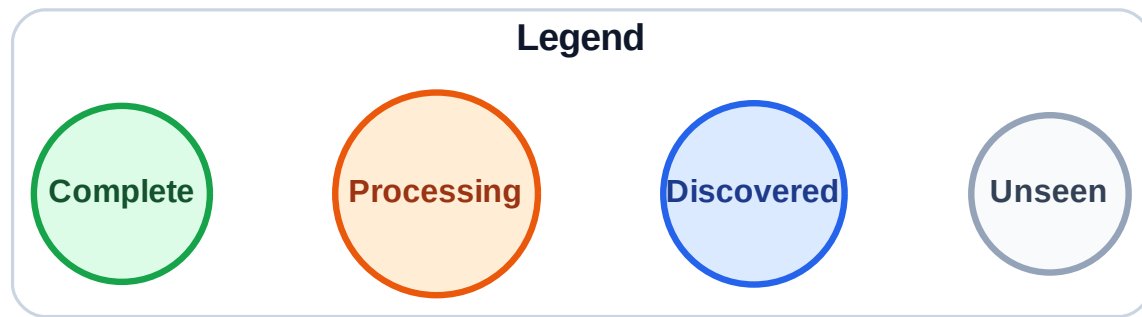
Queue: I → J → K → M → H → G

Visited: A, B, C, D, E, F, L, N, O



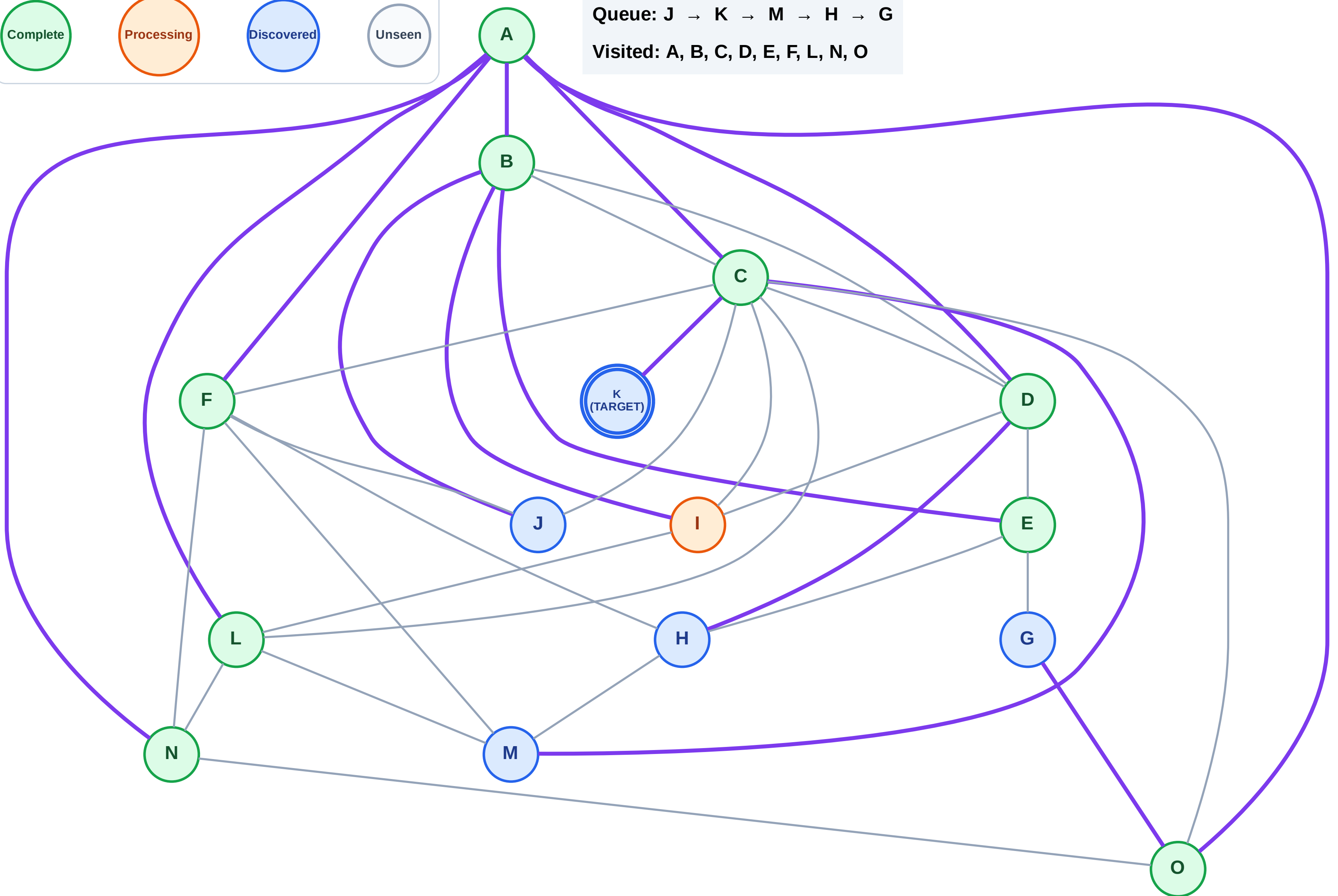
BFS Traversal

Step 20: Process node I



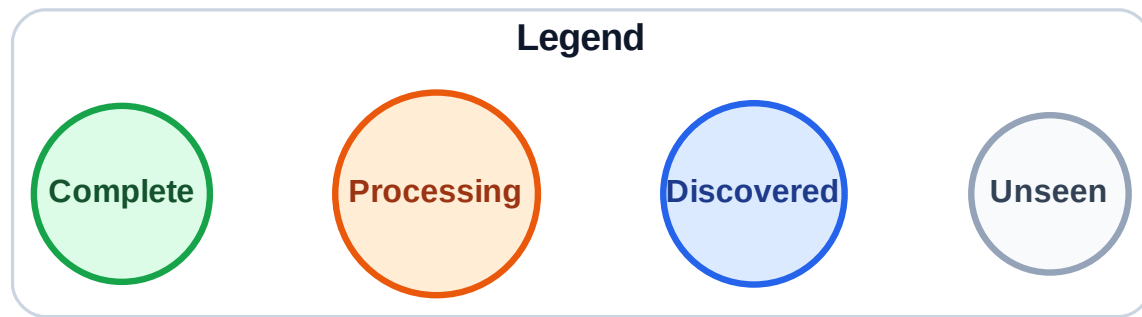
Queue: J → K → M → H → G

Visited: A, B, C, D, E, F, L, N, O



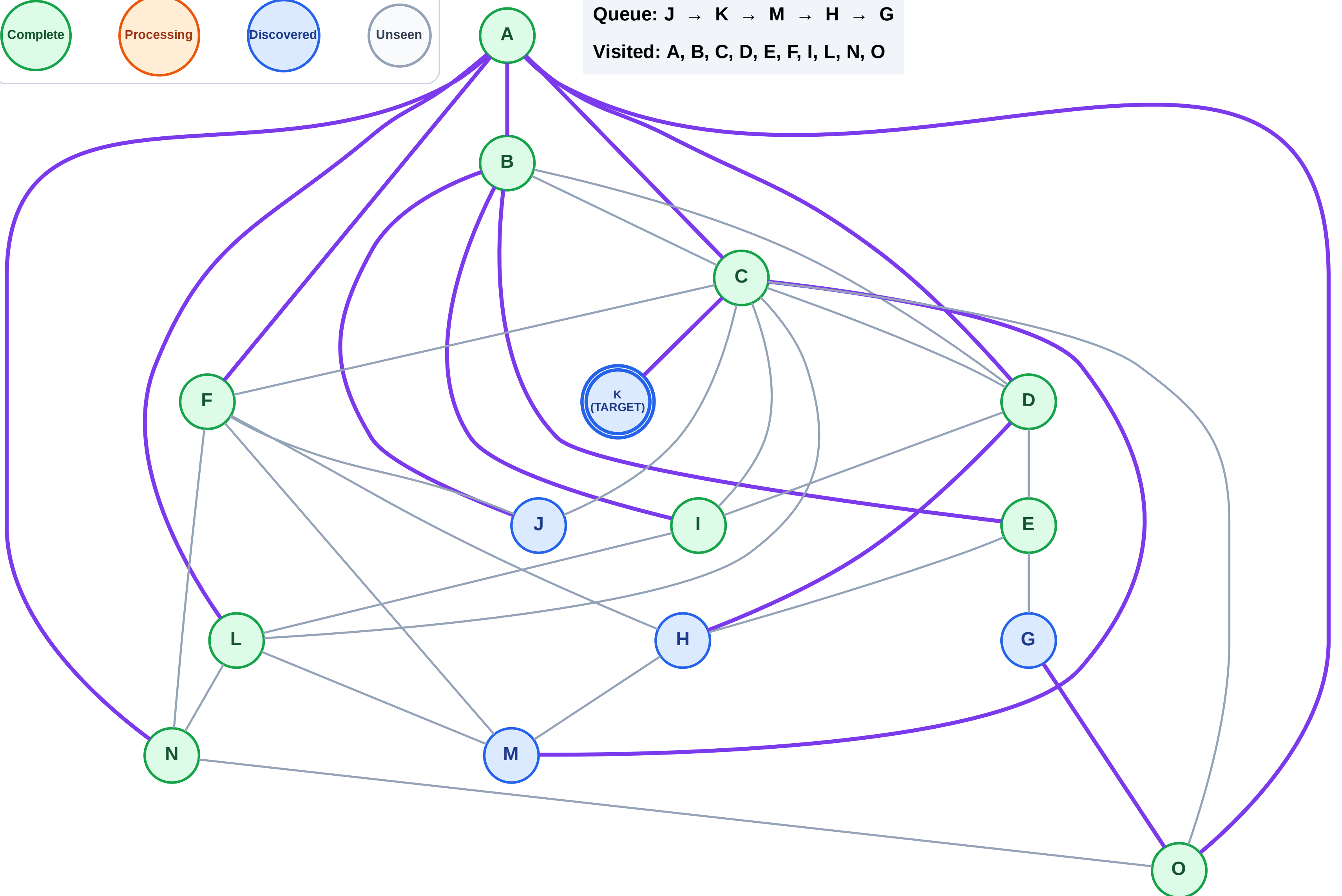
BFS Traversal

Step 21: No new neighbors from I



Queue: J → K → M → H → G

Visited: A, B, C, D, E, F, I, L, N, O



BFS Traversal

Step 22: Process node J

Legend

Complete

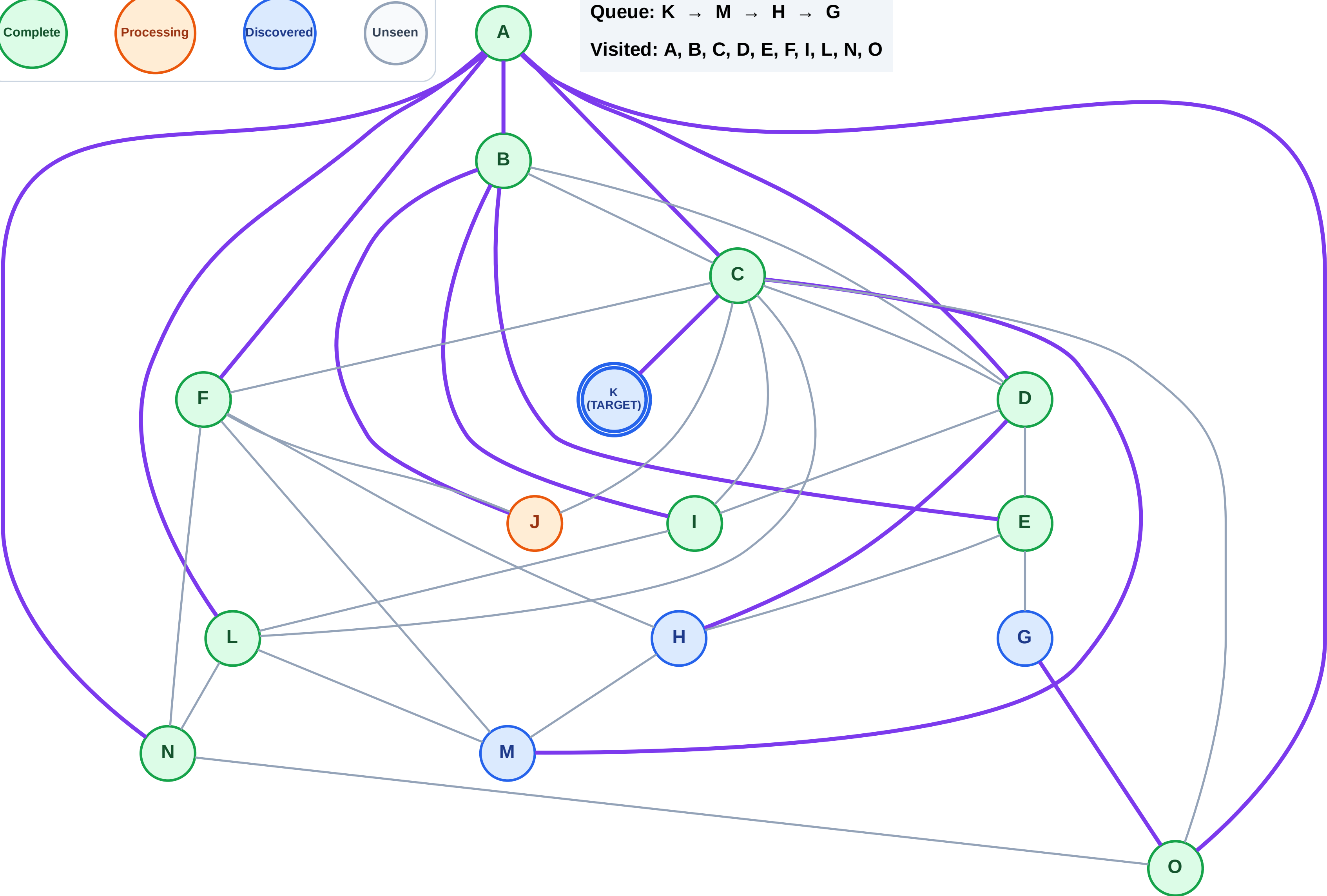
Processing

Discovered

Unseen

Queue: K → M → H → G

Visited: A, B, C, D, E, F, I, L, N, O



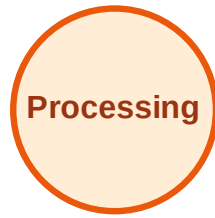
BFS Traversal

Step 23: No new neighbors from J

Legend



Complete



Processing



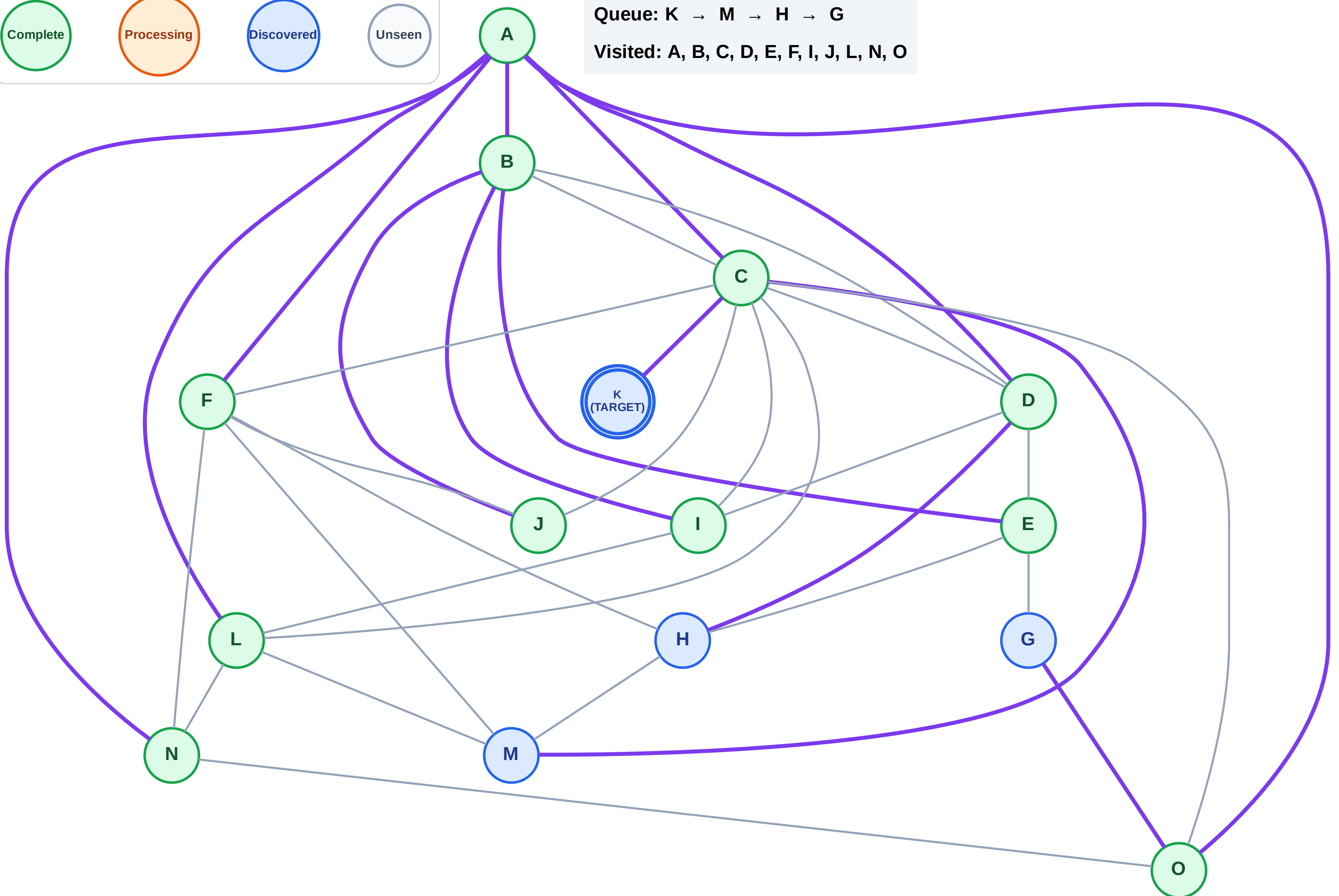
Discovered



Unseen

Queue: K → M → H → G

Visited: A, B, C, D, E, F, I, J, L, N, O



BFS Traversal

Step 24: Process node K

Legend

Complete

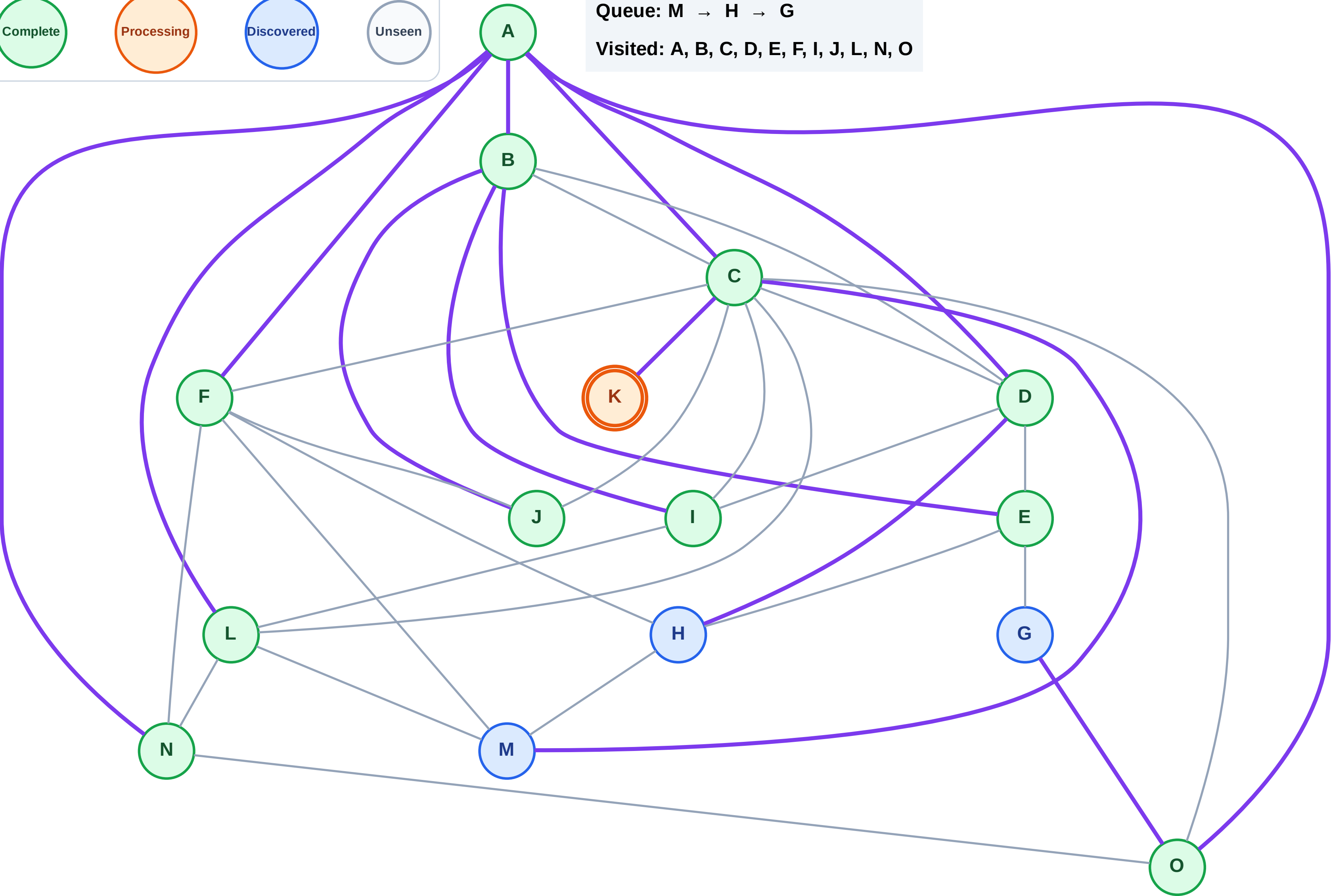
Processing

Discovered

Unseen

Queue: M → H → G

Visited: A, B, C, D, E, F, I, J, L, N, O



BFS Traversal

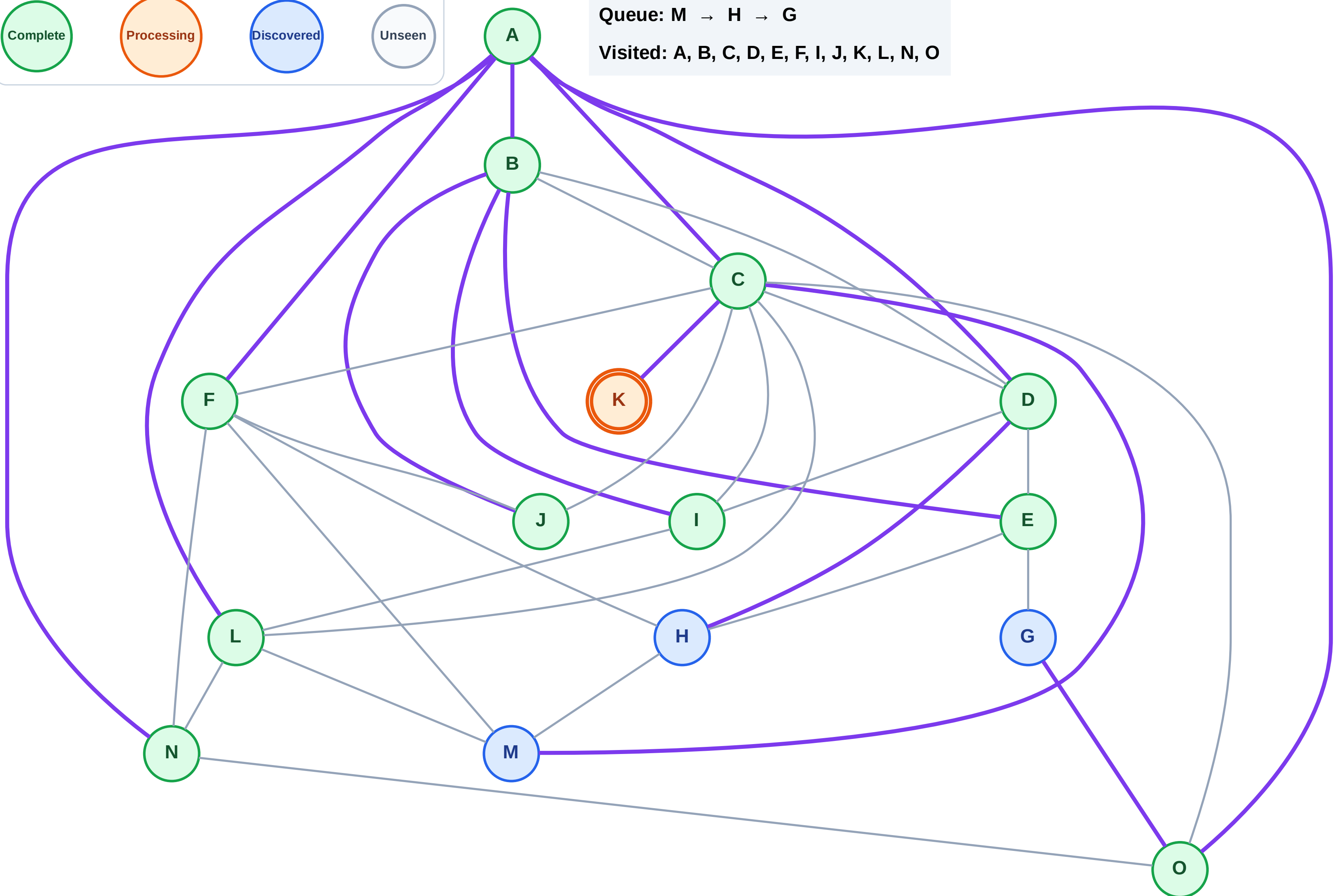
Step 25: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



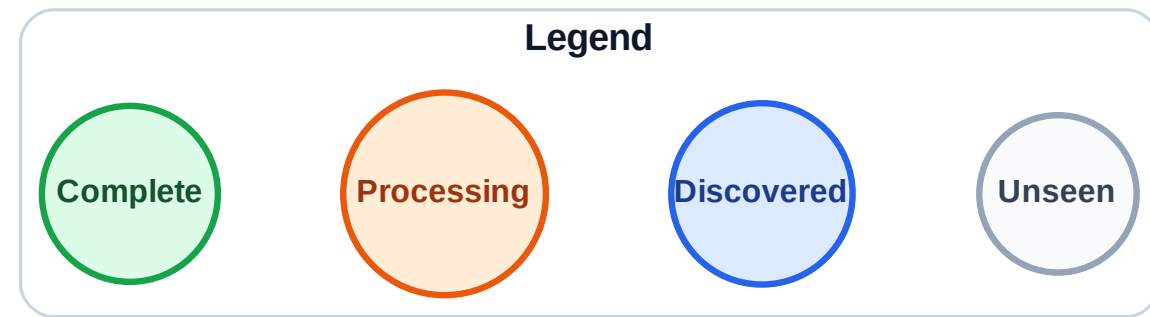
Queue: M → H → G

Visited: A, B, C, D, E, F, I, J, K, L, N, O

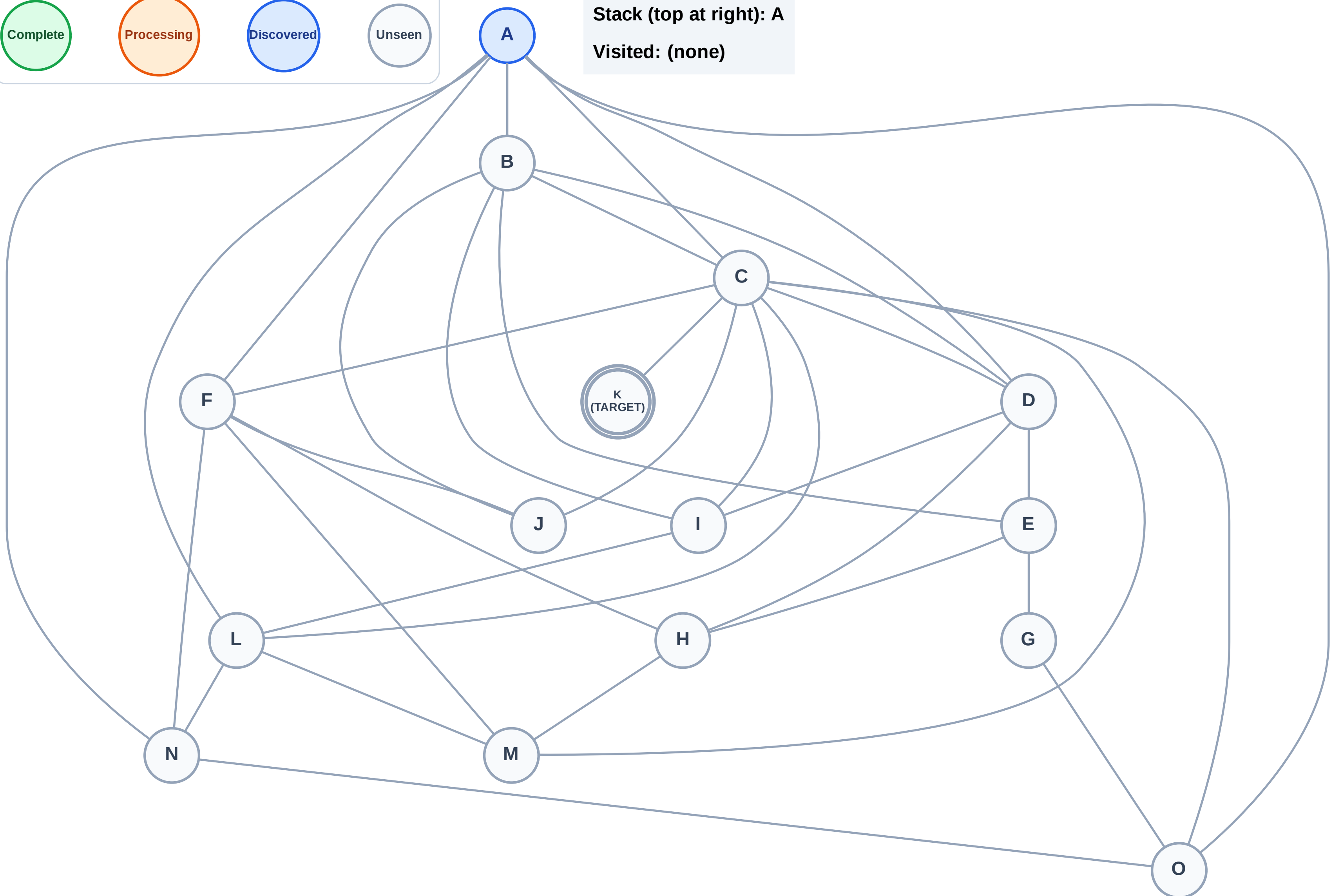


DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A
Visited: (none)



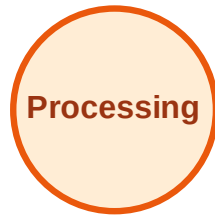
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



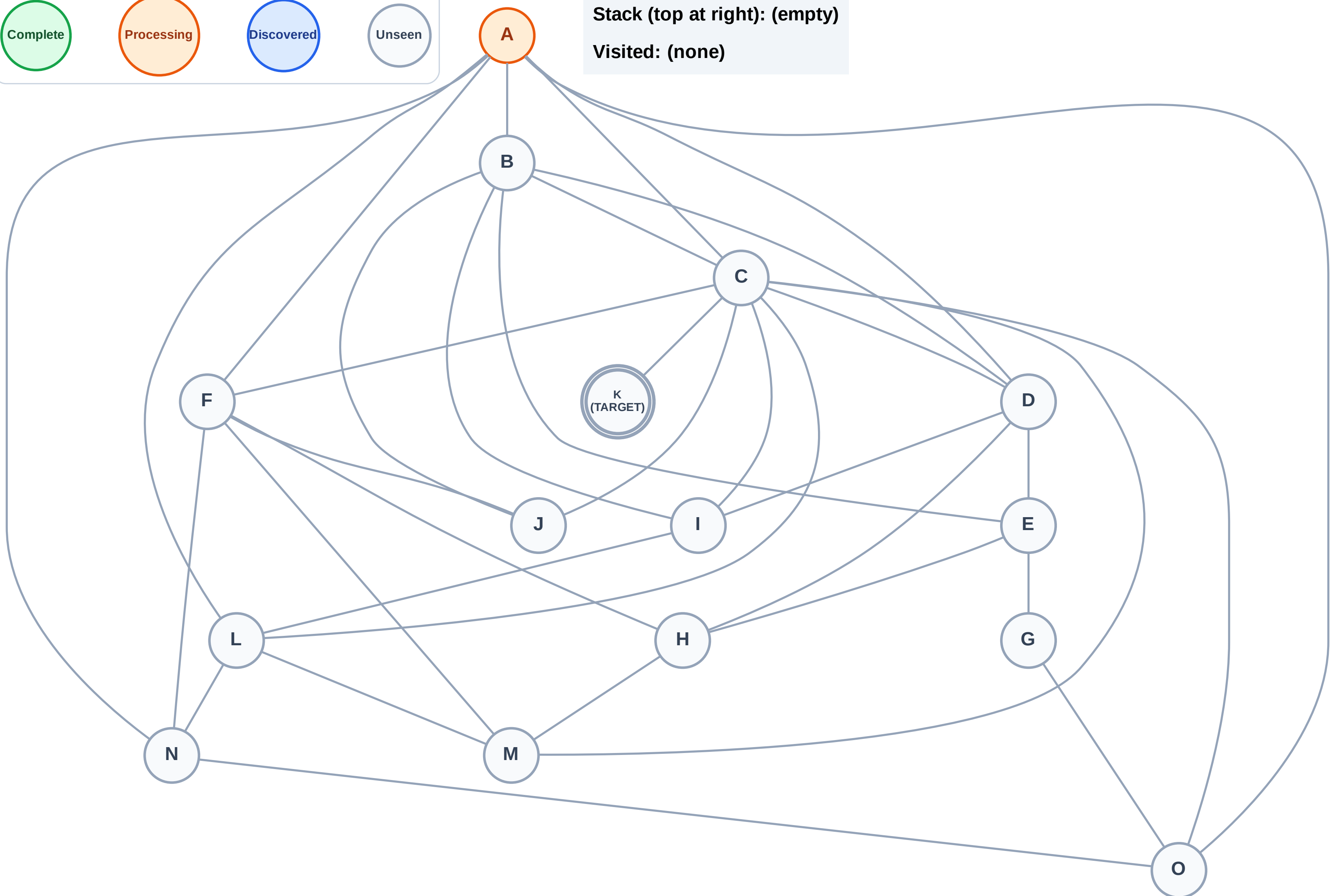
Discovered



Unseen

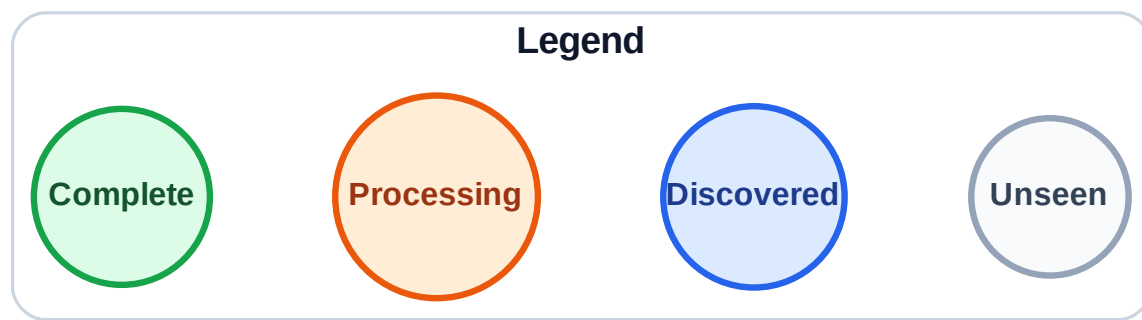
Stack (top at right): (empty)

Visited: (none)



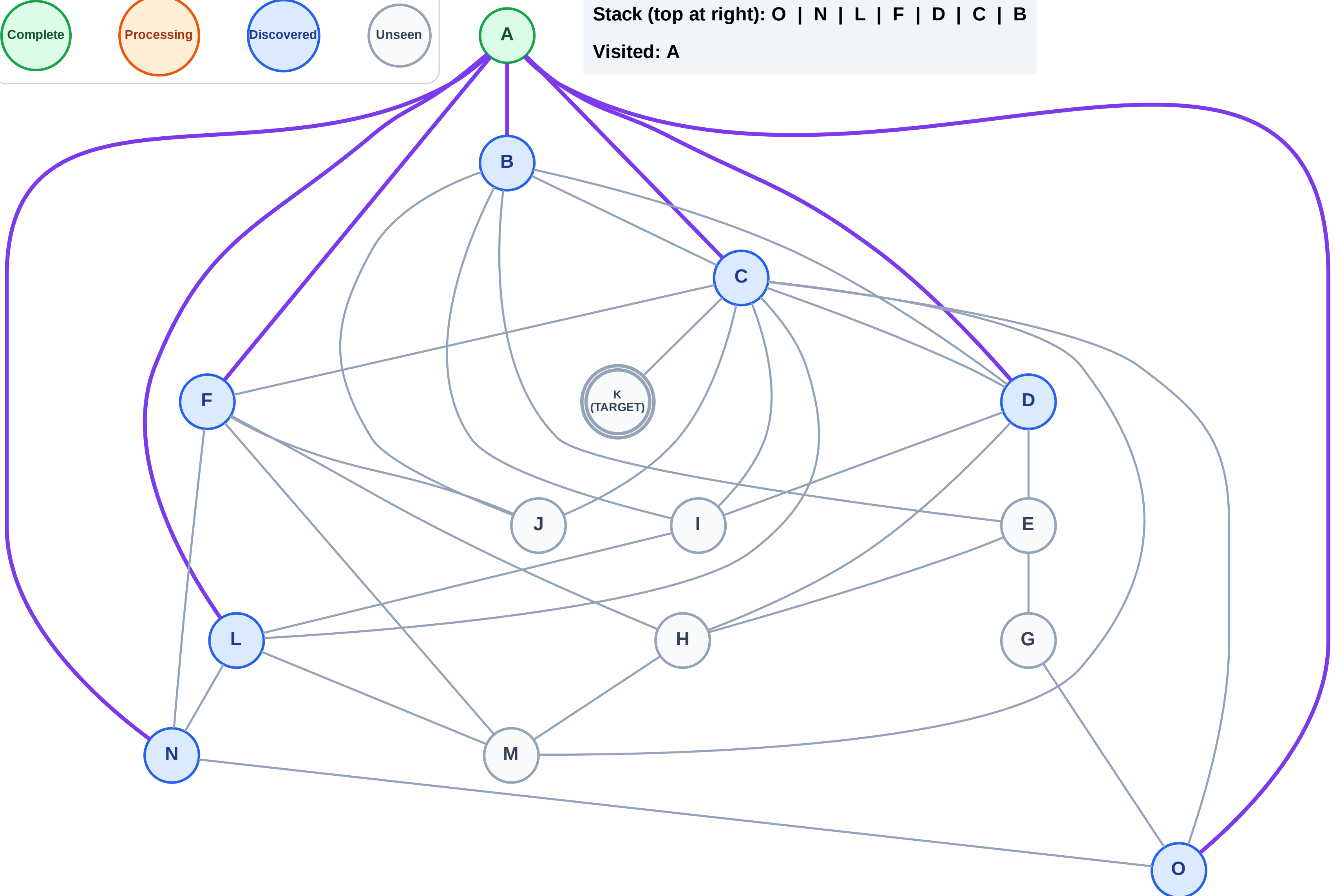
DFS Traversal

Step 3: Discover O, N, L, F, D, C, B from A



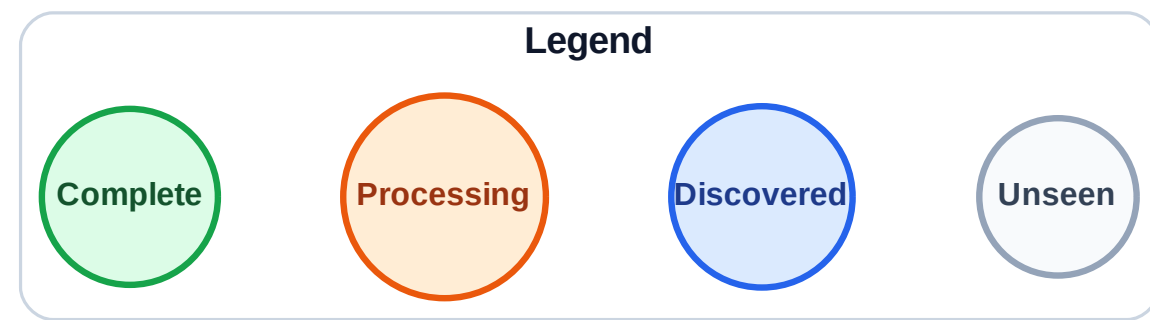
Stack (top at right): O | N | L | F | D | C | B

Visited: A



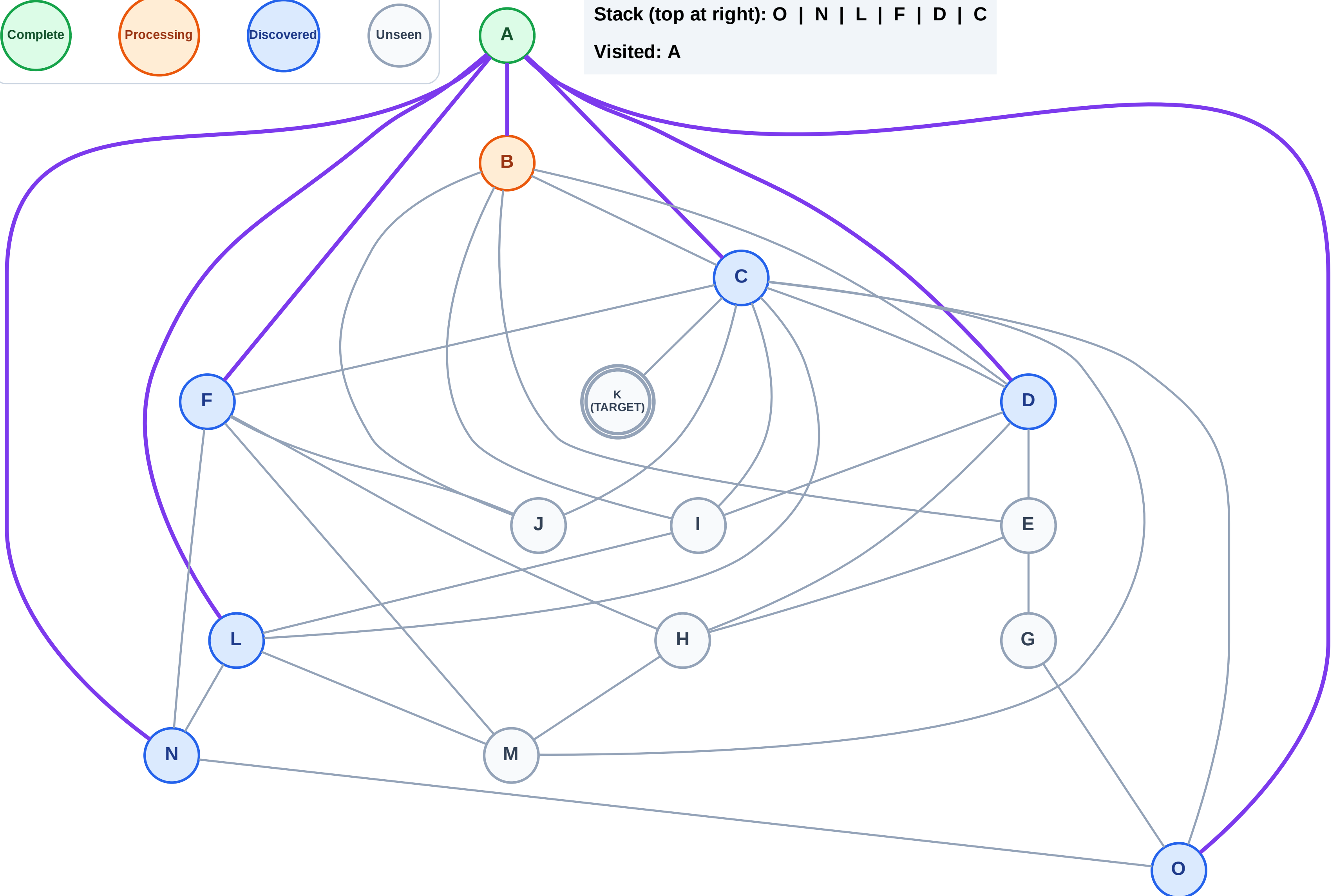
DFS Traversal

Step 4: Process node B



Stack (top at right): O | N | L | F | D | C

Visited: A



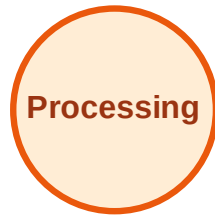
DFS Traversal

Step 5: Discover J, I, E from B

Legend



Complete



Processing



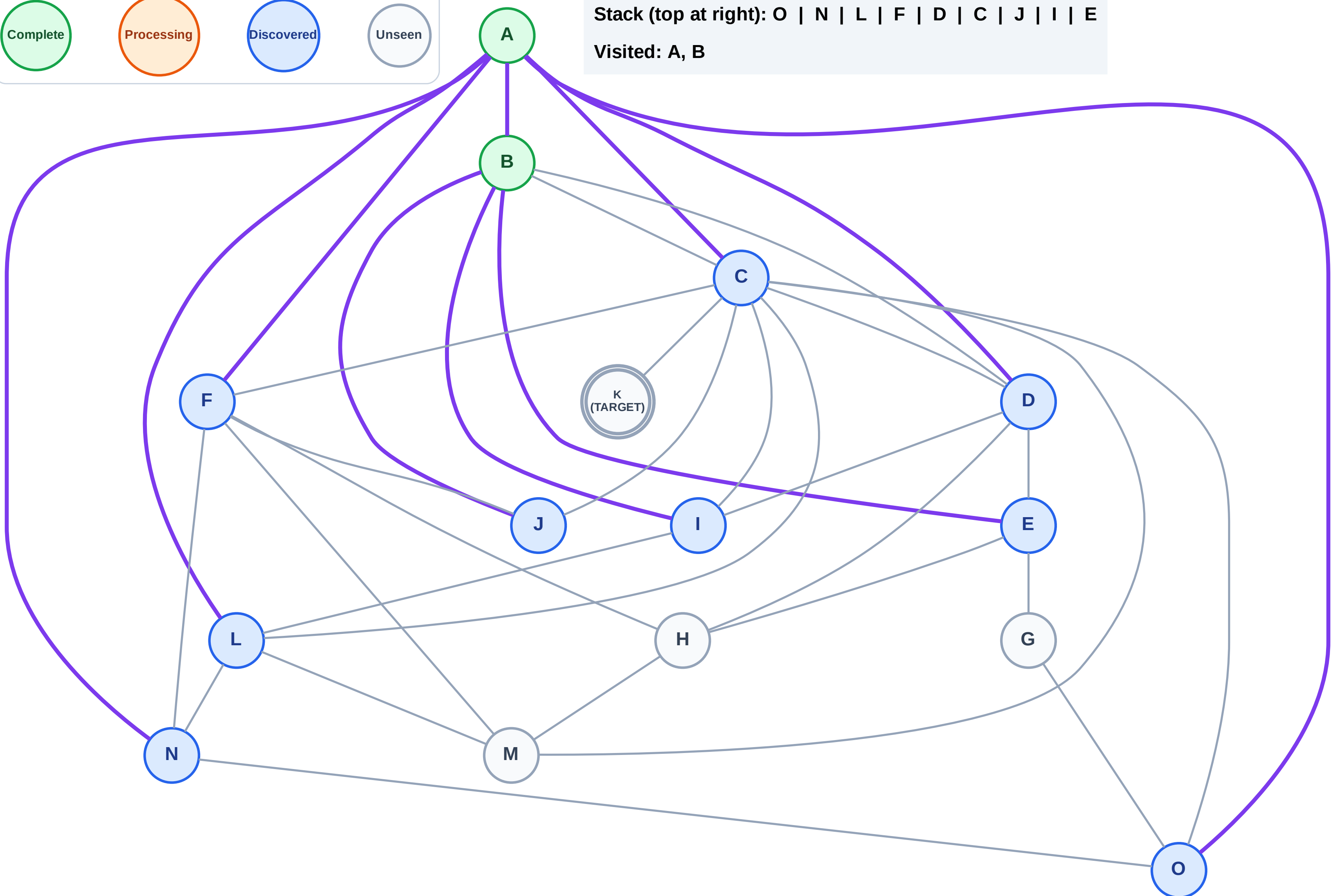
Discovered



Unseen

Stack (top at right): O | N | L | F | D | C | J | I | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

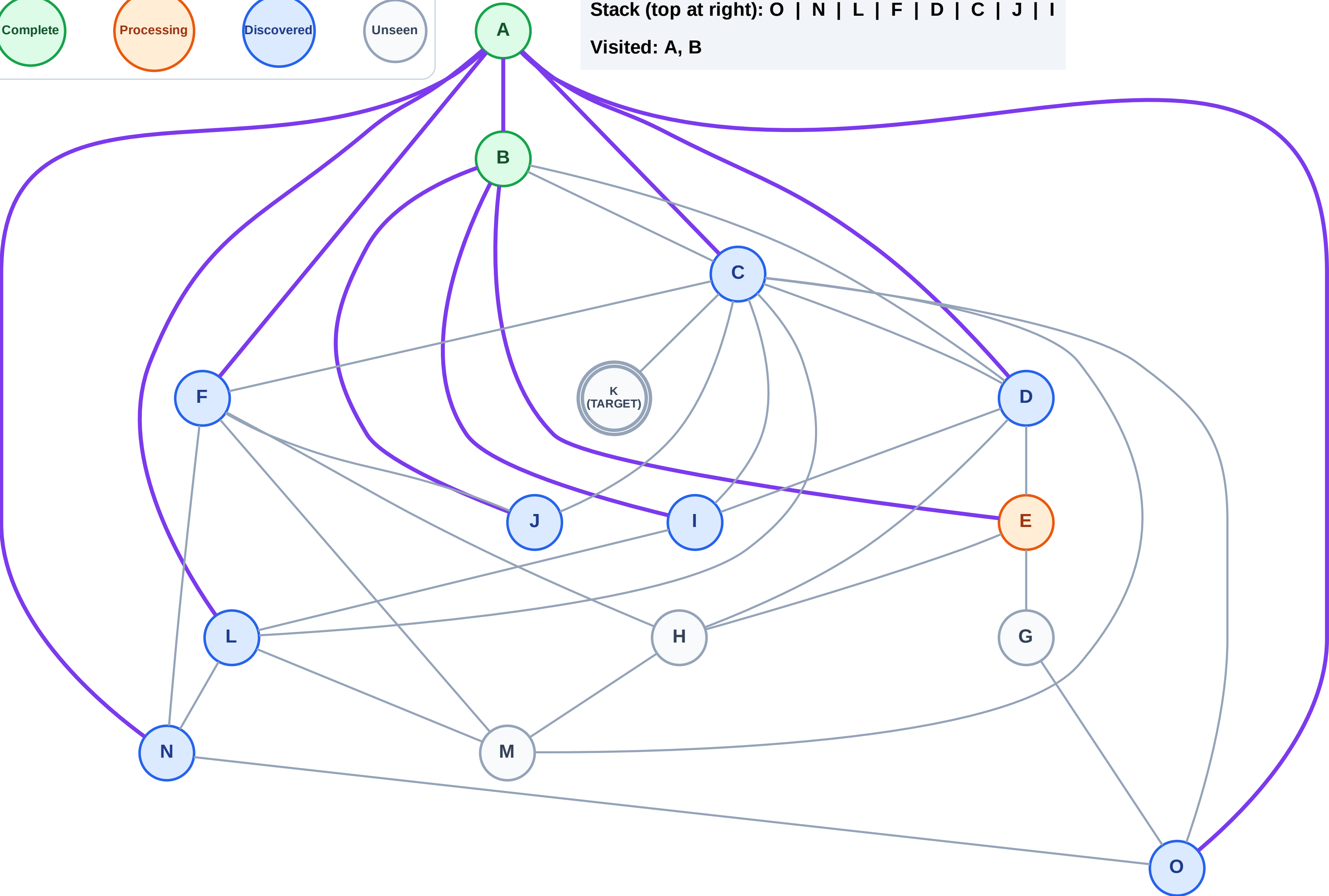
Processing

Discovered

Unseen

Stack (top at right): O | N | L | F | D | C | J | I

Visited: A, B



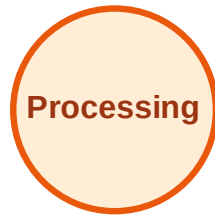
DFS Traversal

Step 7: Discover H, G from E

Legend



Complete



Processing



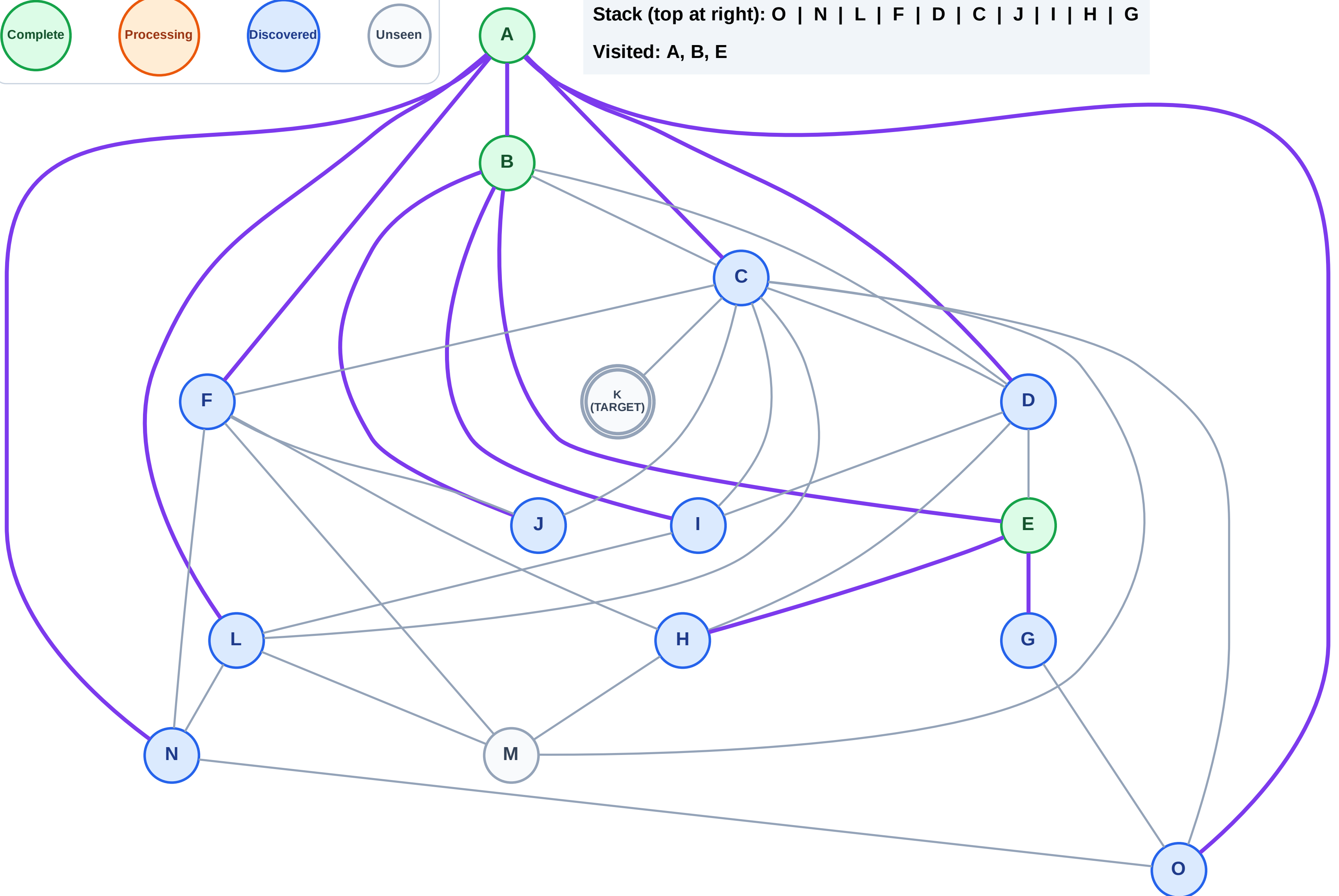
Discovered



Unseen

Stack (top at right): O | N | L | F | D | C | J | I | H | G

Visited: A, B, E



DFS Traversal

Step 8: Process node G

Legend

Complete

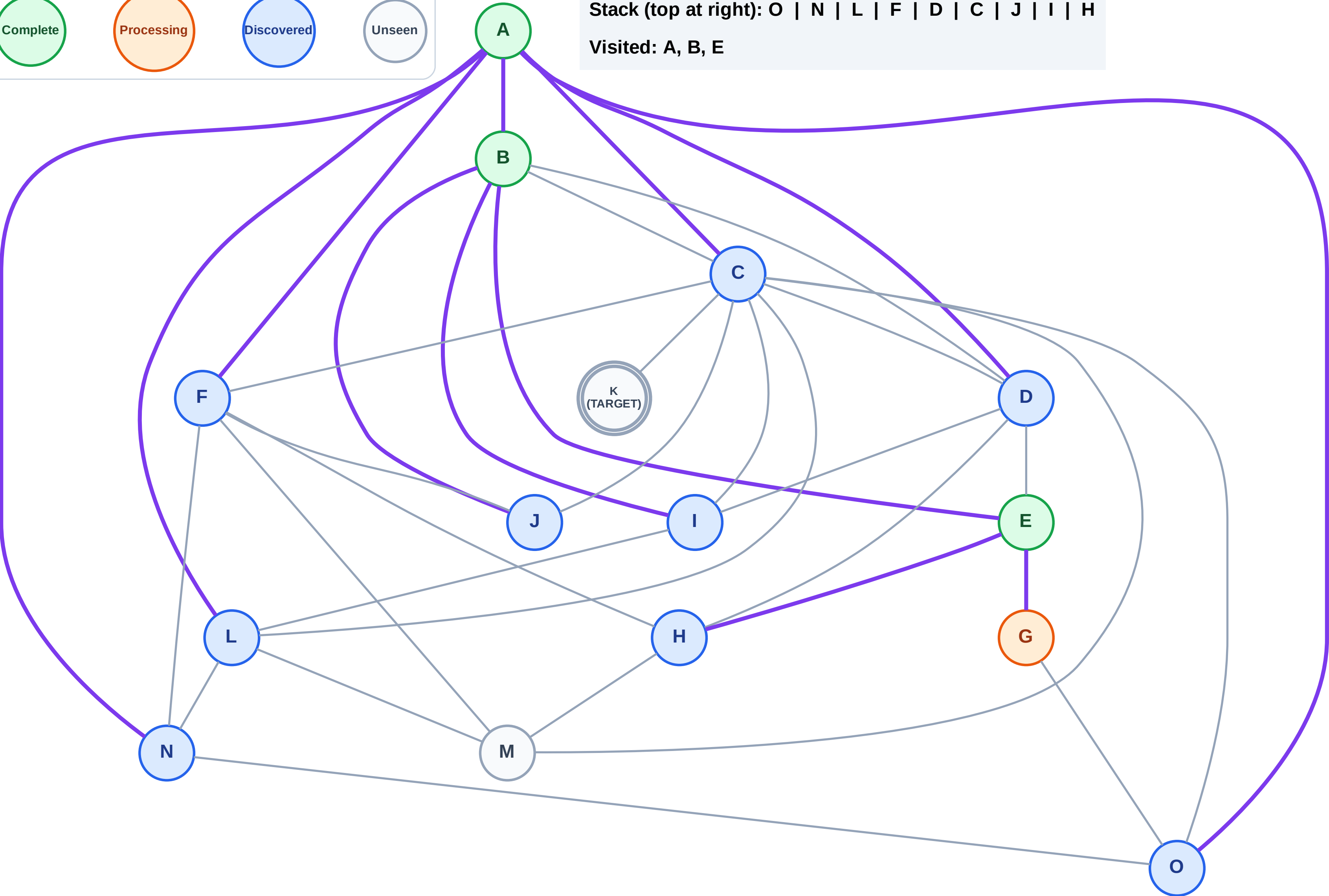
Processing

Discovered

Unseen

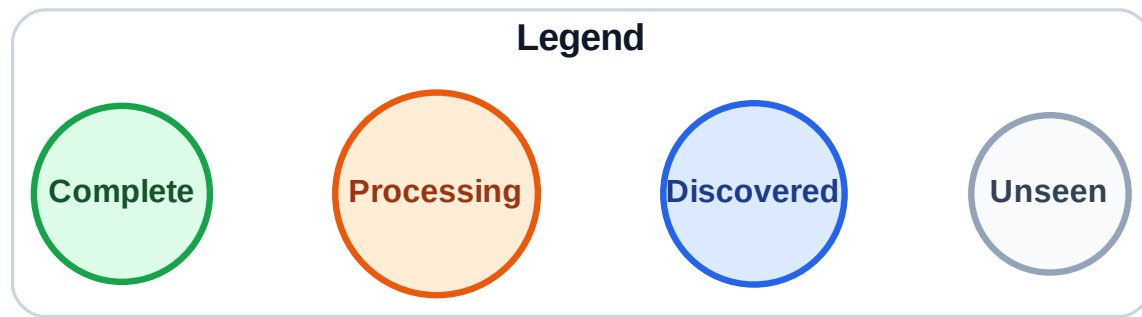
Stack (top at right): O | N | L | F | D | C | J | I | H

Visited: A, B, E



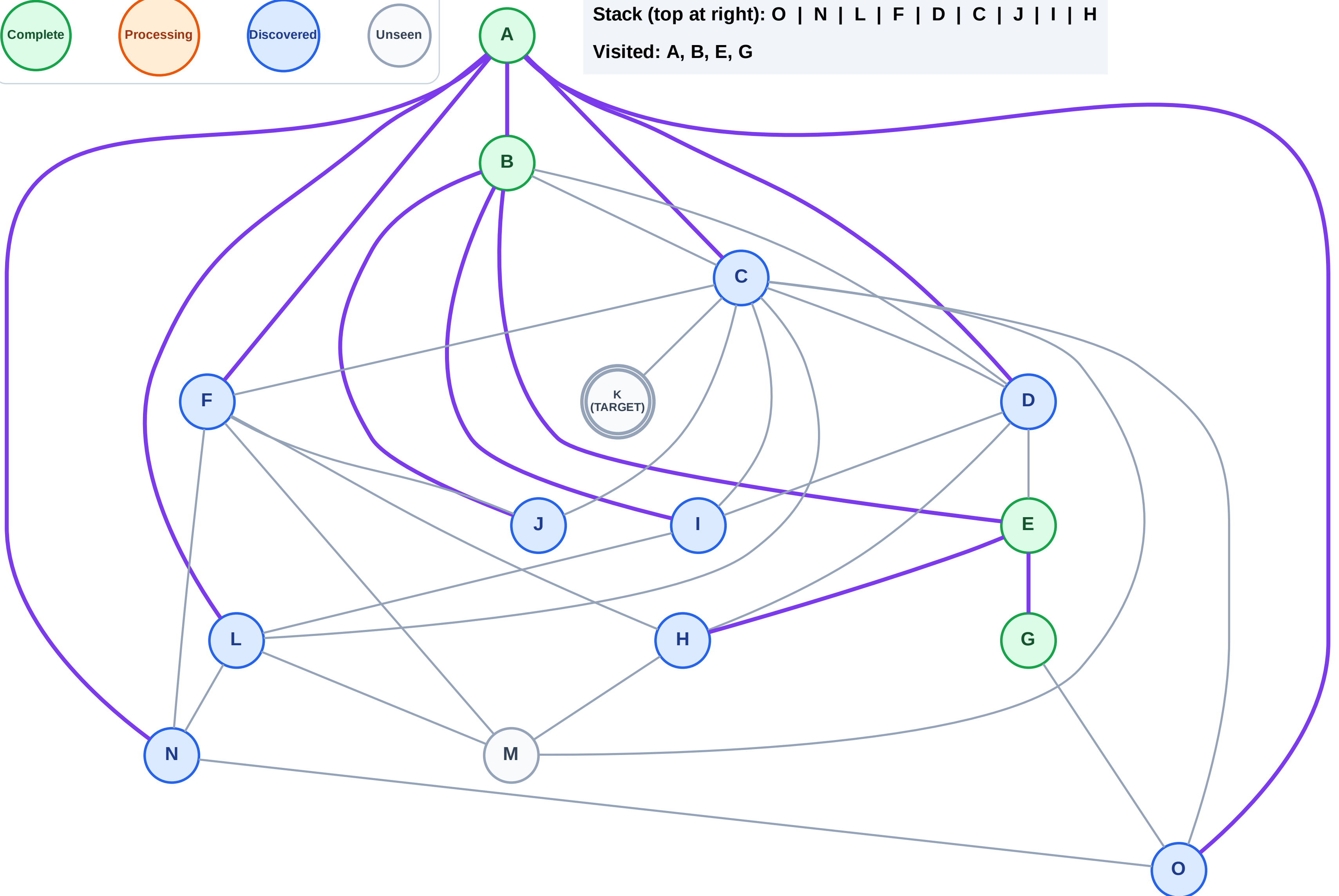
DFS Traversal

Step 9: No new neighbors from G



Stack (top at right): O | N | L | F | D | C | J | I | H

Visited: A, B, E, G



DFS Traversal

Step 10: Process node H

Legend

Complete

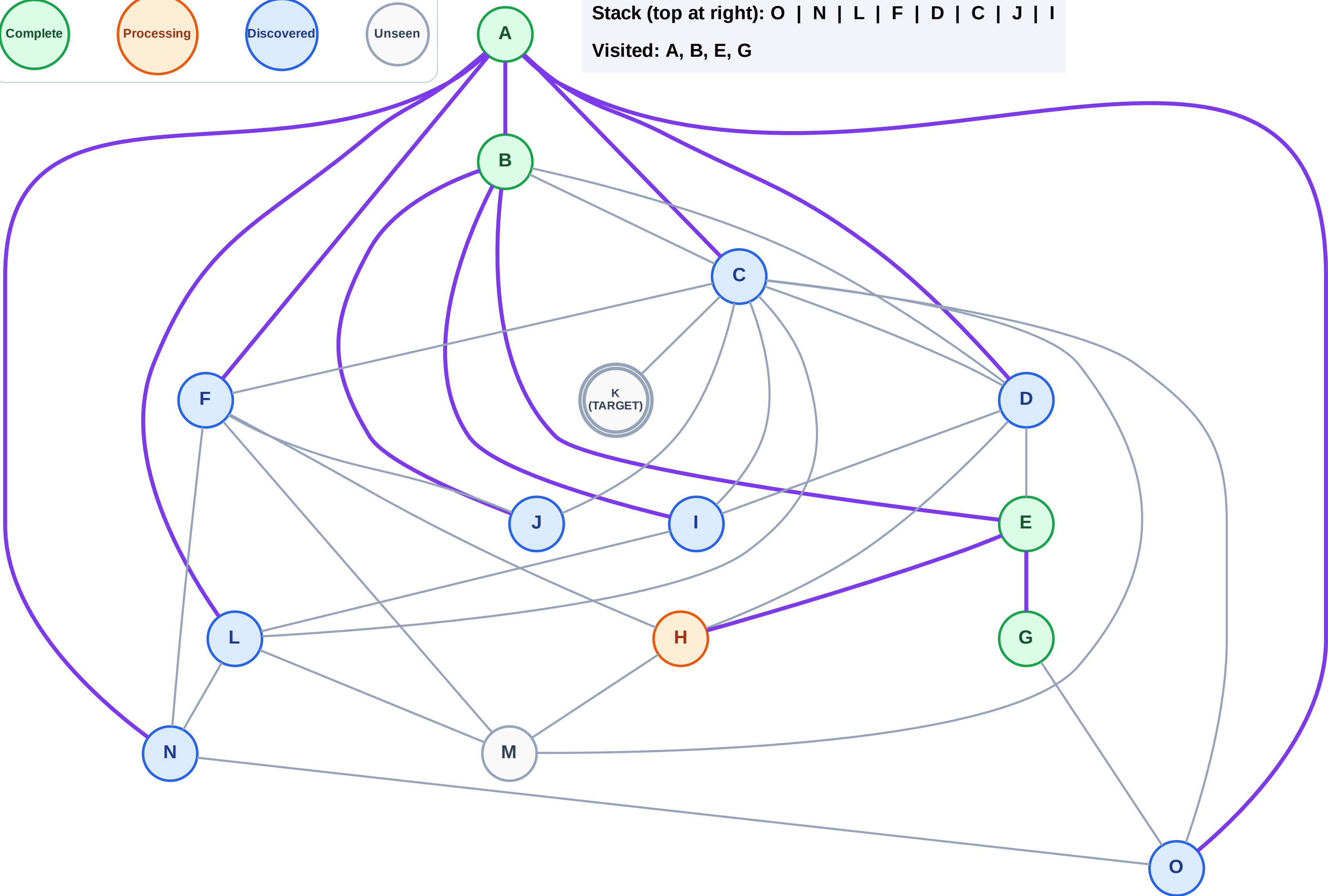
Processing

Discovered

Unseen

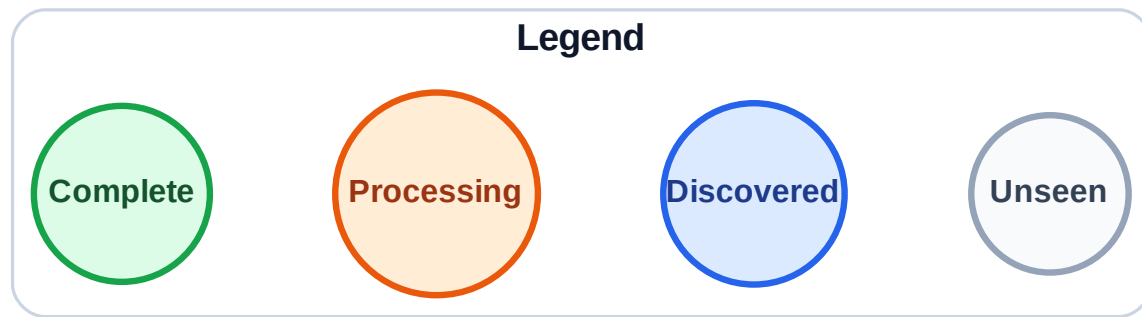
Stack (top at right): O | N | L | F | D | C | J | I

Visited: A, B, E, G



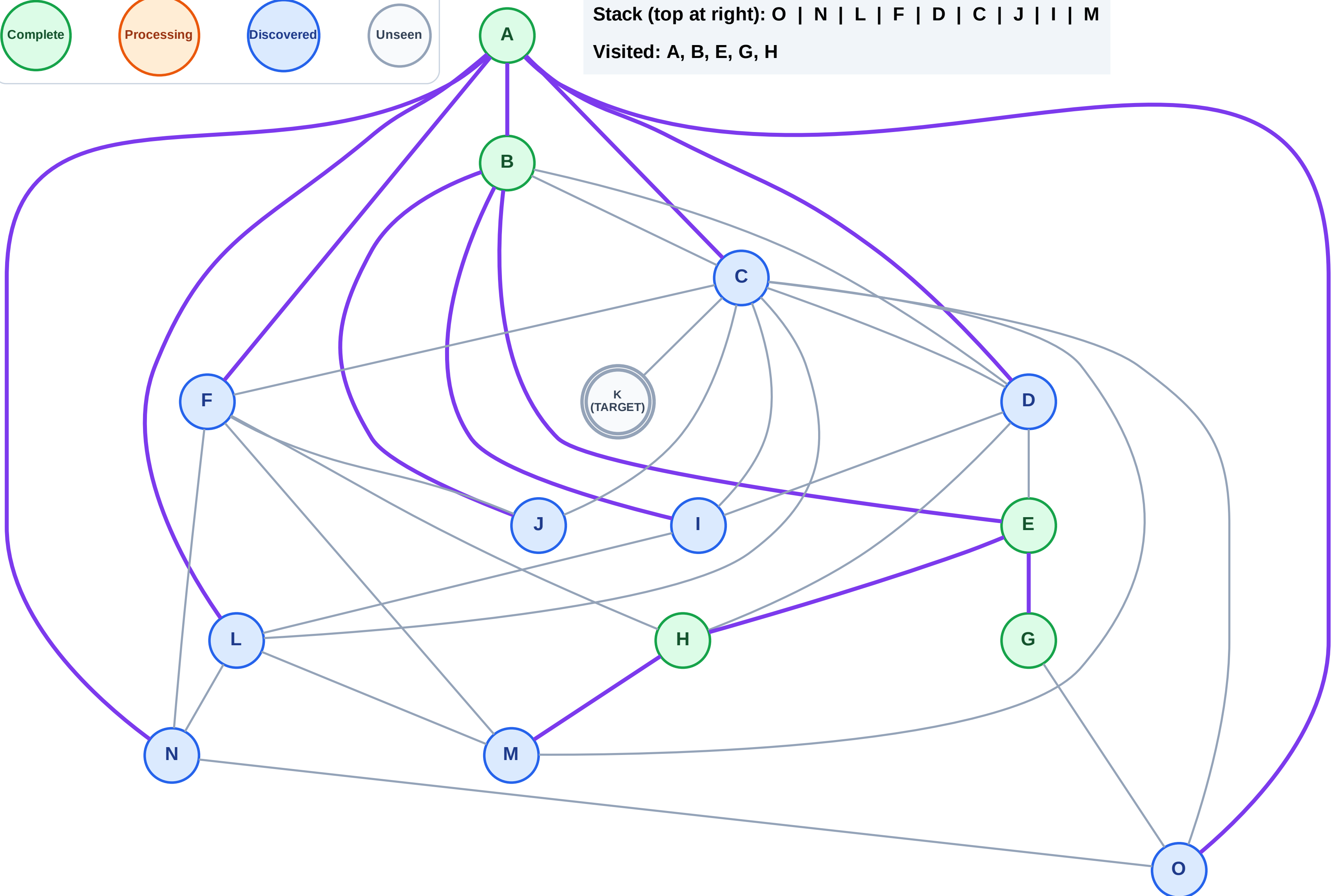
DFS Traversal

Step 11: Discover M from H



Stack (top at right): O | N | L | F | D | C | J | I | M

Visited: A, B, E, G, H



DFS Traversal

Step 12: Process node M

Legend

Complete

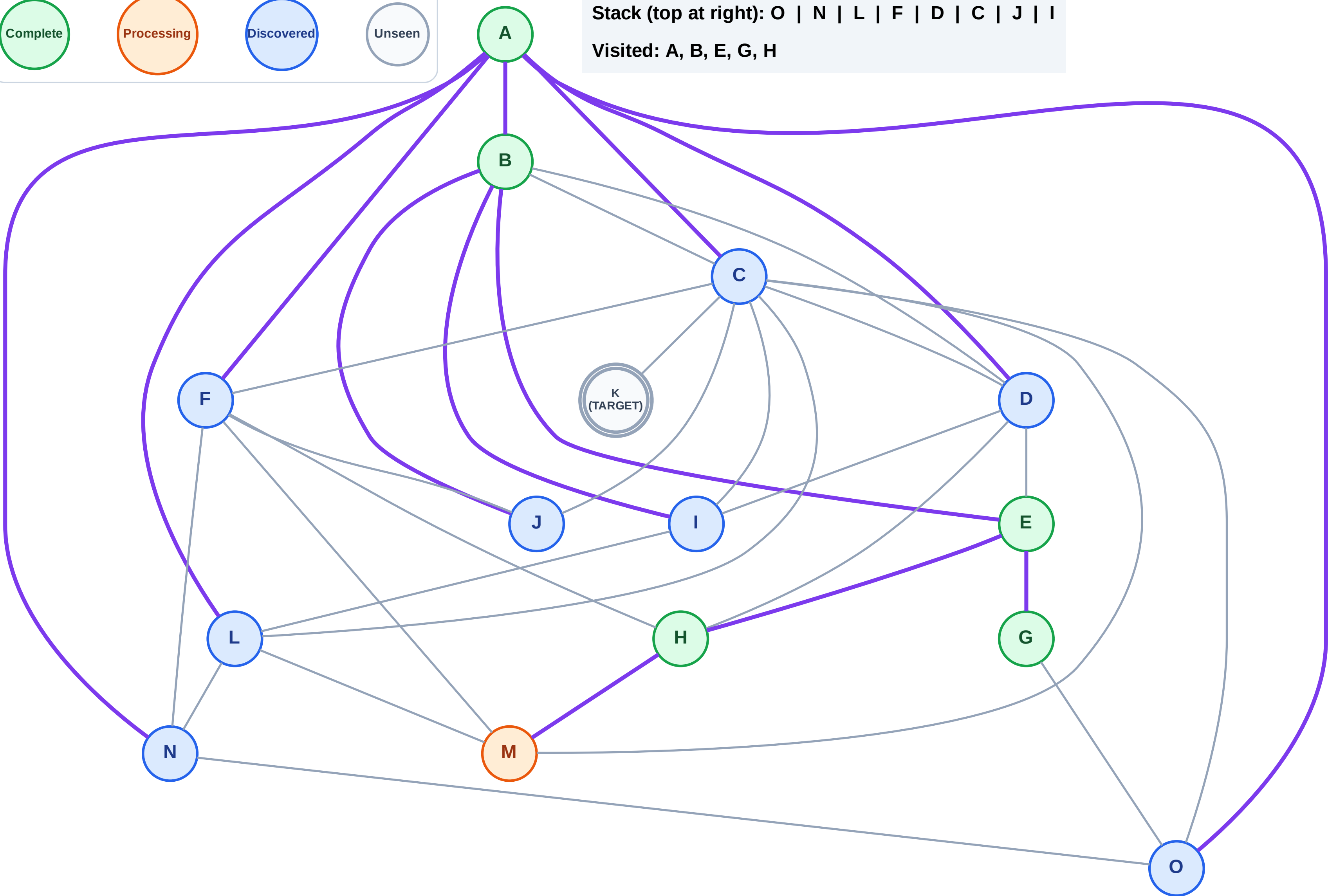
Processing

Discovered

Unseen

Stack (top at right): O | N | L | F | D | C | J | I

Visited: A, B, E, G, H



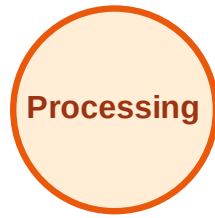
DFS Traversal

Step 13: No new neighbors from M

Legend



Complete



Processing



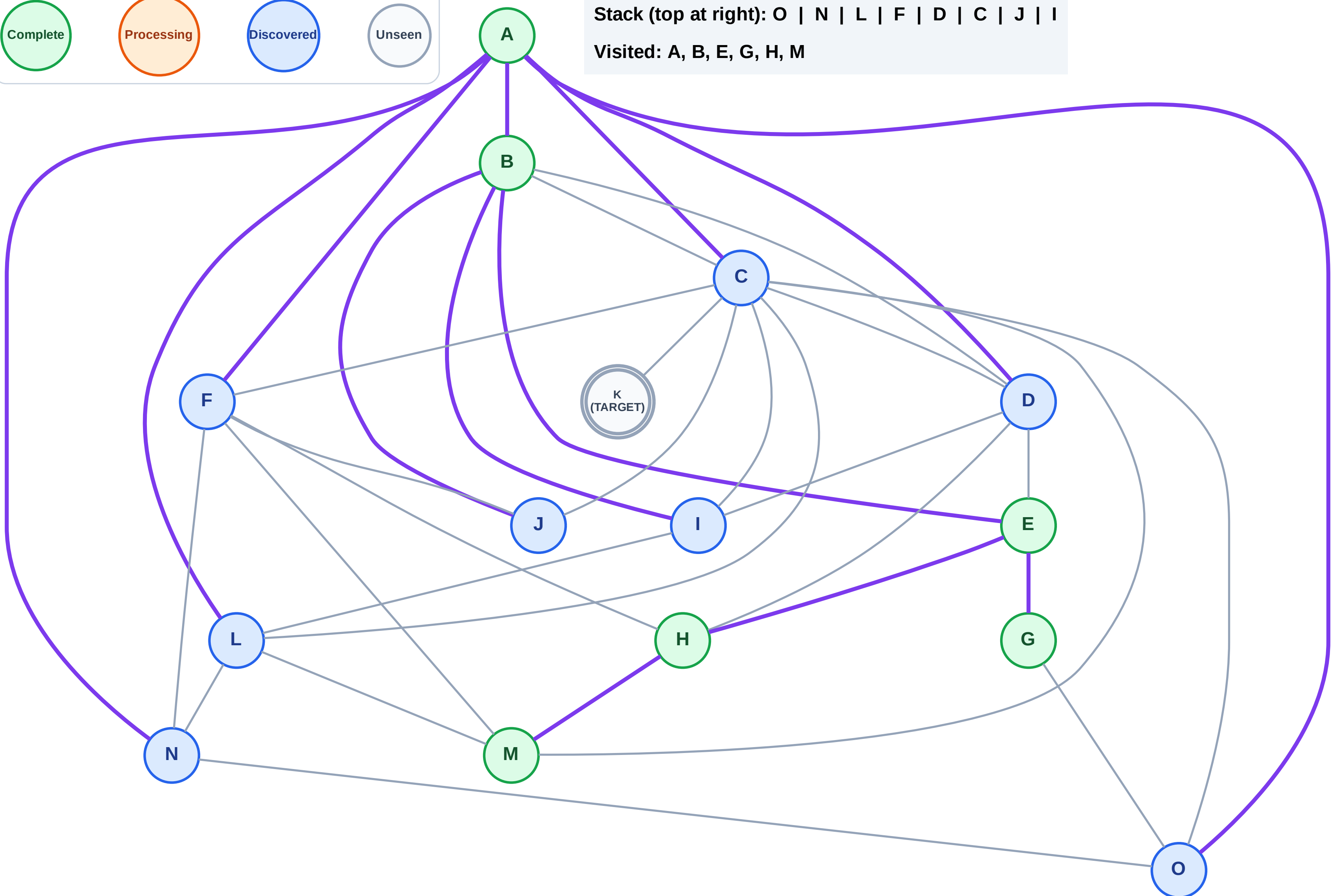
Discovered



Unseen

Stack (top at right): O | N | L | F | D | C | J | I

Visited: A, B, E, G, H, M



DFS Traversal

Step 14: Process node I

Legend

Complete

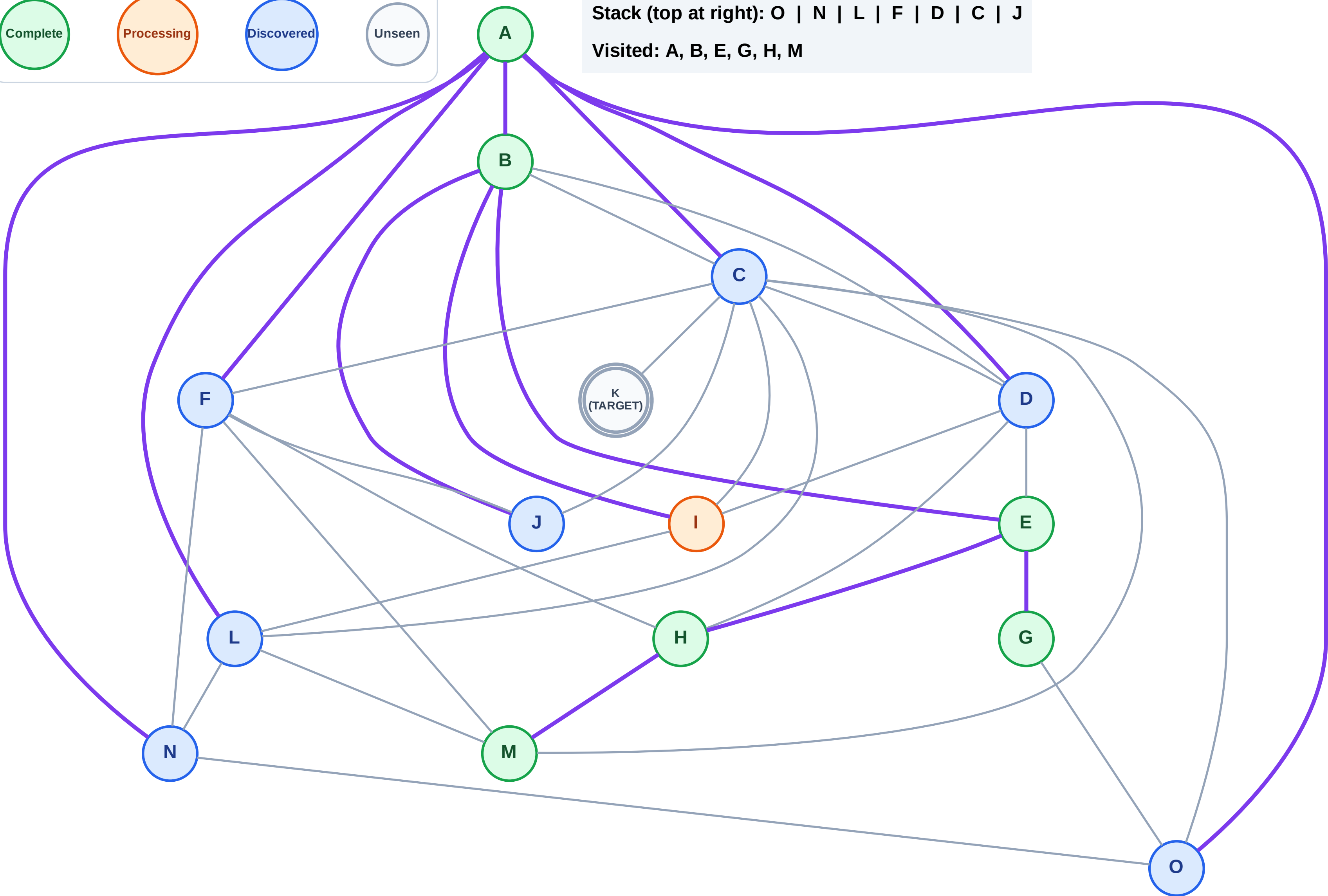
Processing

Discovered

Unseen

Stack (top at right): O | N | L | F | D | C | J

Visited: A, B, E, G, H, M



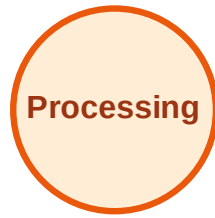
DFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



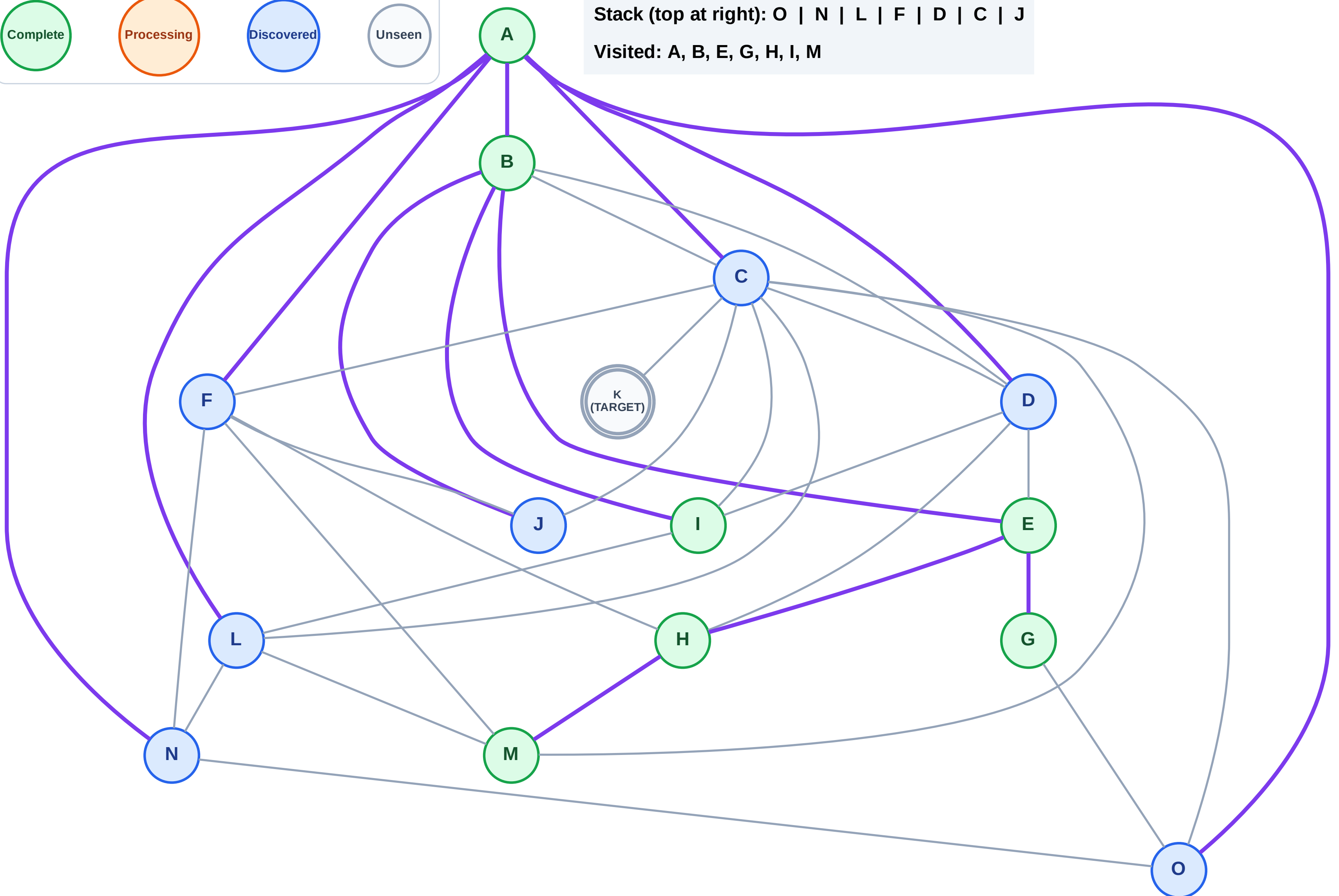
Discovered



Unseen

Stack (top at right): O | N | L | F | D | C | J

Visited: A, B, E, G, H, I, M



DFS Traversal

Step 16: Process node J

Legend

Complete

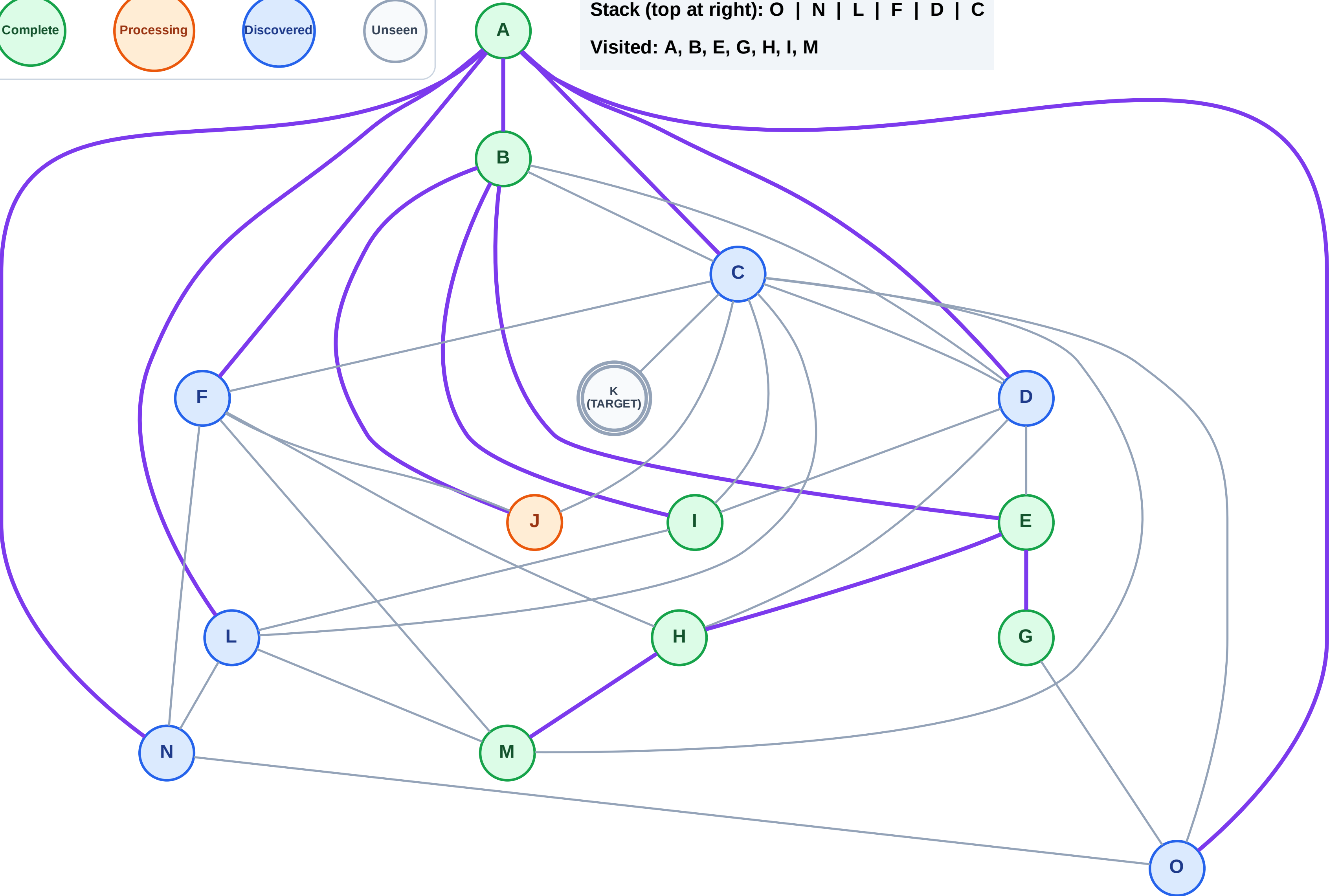
Processing

Discovered

Unseen

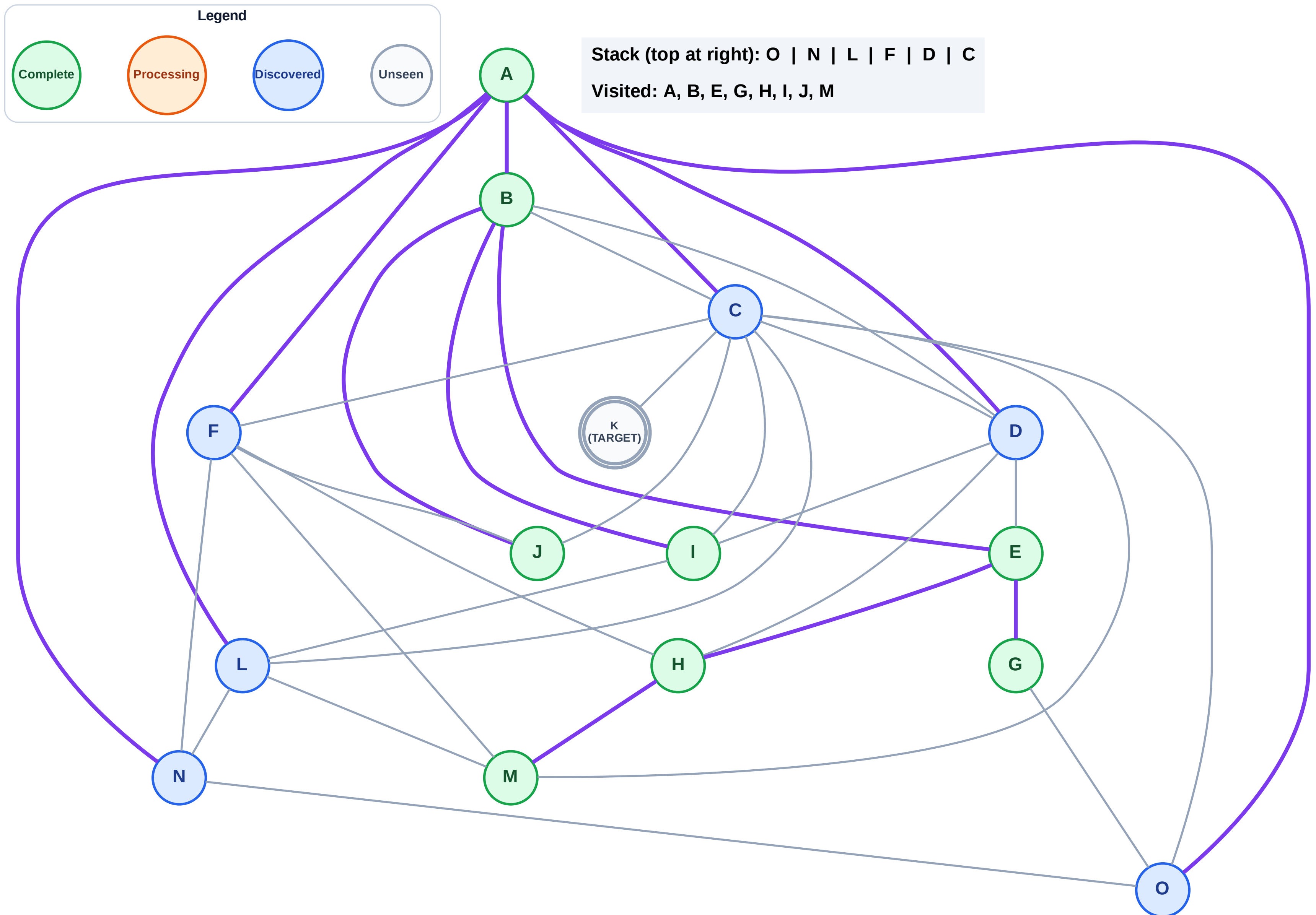
Stack (top at right): O | N | L | F | D | C

Visited: A, B, E, G, H, I, M



Step 17: No new neighbors from J

Step 17: No new neighbors from J



DFS Traversal

Step 18: Process node C

Legend

Complete

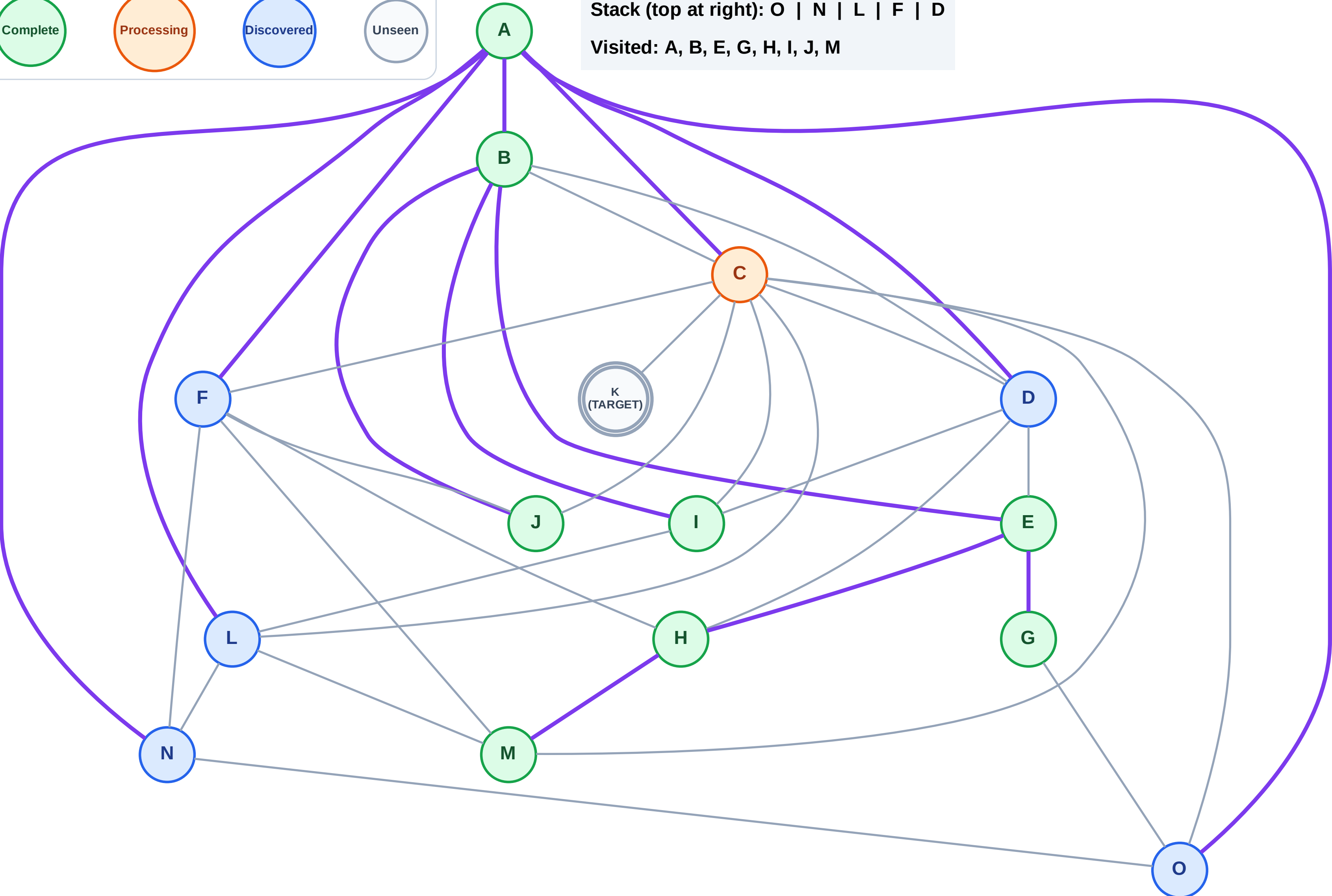
Processing

Discovered

Unseen

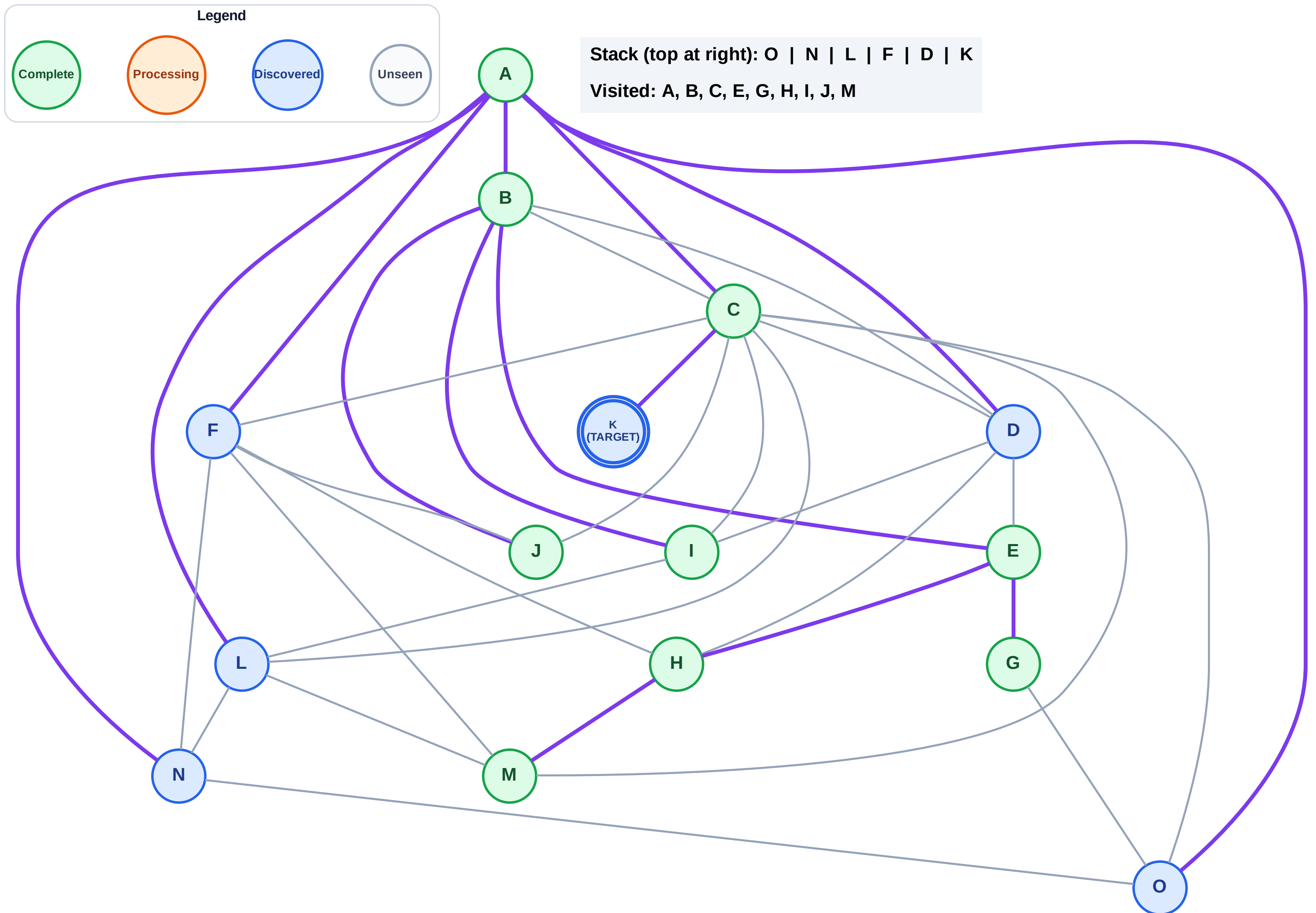
Stack (top at right): O | N | L | F | D

Visited: A, B, E, G, H, I, J, M



Step 19: Discover K from C

Step 19: Discover K from C



DFS Traversal

Step 20: Process node K

Legend

Complete

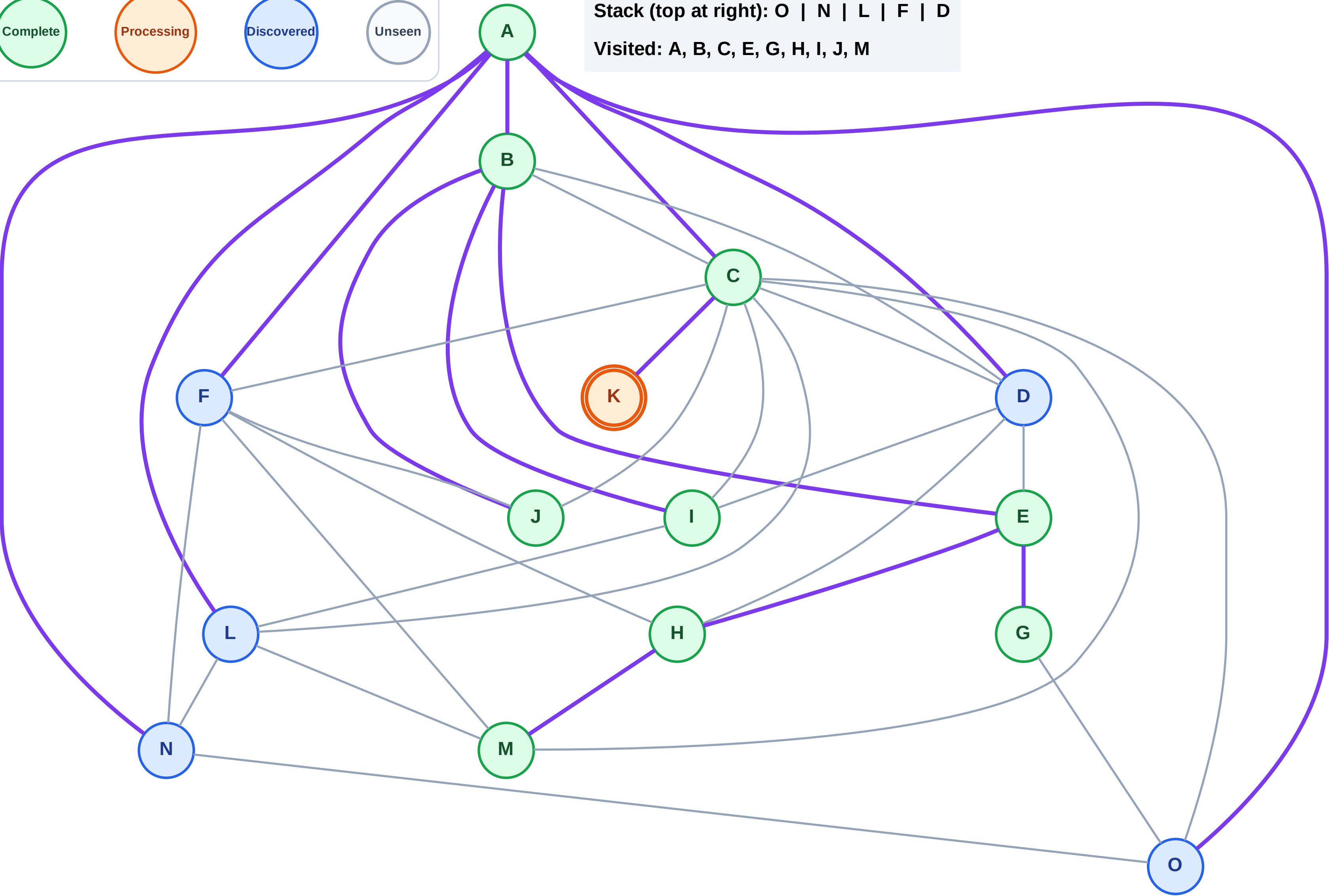
Processing

Discovered

Unseen

Stack (top at right): O | N | L | F | D

Visited: A, B, C, E, G, H, I, J, M



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | L | F | D

Visited: A, B, C, E, G, H, I, J, K, M

